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The 23rd International Conference on Manufacturing Research (ICMR2026)

Conference Programme

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WELCOME MESSAGE

Dear Distinguished Guests, Keynote Speakers, Authors, Delegates, and Colleagues,

It is my great pleasure to welcome you to the **23rd International Conference on Manufacturing Research (ICMR 2026)**, hosted at **Nazarbayev University, Astana, Kazakhstan, from 2–4 September 2026**.

ICMR has long served as an international platform for researchers, academics, industry professionals, and practitioners to exchange knowledge, present new developments, and discuss emerging challenges in manufacturing. We are honored to continue this tradition and to host ICMR 2026 at Nazarbayev University in Kazakhstan.

This year's conference brings together research covering a wide range of topics, including advanced manufacturing, additive manufacturing, artificial intelligence and machine learning, digital and smart manufacturing, robotics and automation, advanced materials, sustainable manufacturing, and Industry 4.0. The technical program, keynote presentations, and parallel sessions provide valuable opportunities for sharing ideas, developing collaborations, and strengthening links between academia and industry.

On behalf of the Organizing Committee, I would like to sincerely thank all authors, reviewers, keynote and invited speakers, session chairs, committee members, partners, volunteers, and supporting staff for their valuable contributions and commitment to the success of the conference.

We are also delighted to welcome all participants to **Astana**, a modern and dynamic city known for its distinctive architecture, hospitality, and growing role as a center for education, research, and innovation.

I hope that ICMR 2026 will provide an inspiring and rewarding experience, encourage productive discussions, create new professional connections, and open opportunities for future research and collaboration.

Welcome to ICMR 2026, Nazarbayev University, and Astana, Kazakhstan.

I wish you all a successful, enjoyable, and memorable conference.

Prof. Essam Shehab, General Chair

23rd International Conference on Manufacturing Research (ICMR 2026)
Nazarbayev University, Astana, Kazakhstan

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PLENARY SPEAKER

Prof. Ronan Daly
Institute for Manufacturing, Cambridge University, UK

Biography: Ronan Daly is Professor of Advanced Manufacturing in the Institute for Manufacturing, Department of Engineering, University of Cambridge. He leads the Fluids in Advanced Manufacturing research group, which tackles fundamental research into fluids, interfaces, biomaterials, nanomaterials and chemical systems to support the creation of future device technologies and the advanced manufacturing techniques that make them. This means the team links Technology Push and Pull, thinking about the influence of manufacturing on fundamental research questions, and pioneering the science of scale-up for healthcare and additive manufacturing applications. Ronan is also co-director of the Institute for Biomedical Innovation, University of Cambridge, supporting the translation of new medical technologies towards manufacturing.

<u>**Plenary Session:** Exploring the Role of Academic Manufacturing Research in Delivering New Medical Technologies</u>

Abstract: The Fluids in Advanced Manufacturing research group at Institute for Manufacturing, Department of Engineering, University of Cambridge identifies and explores underpinning scientific challenges in a range of real-world scenarios and manufacturing techniques. Research crosses multiple scales including molecular dynamics, surface science, formulation, stability, micro-scale patterning, and production processes. This presentation will firstly describe the positioning of our academic research within a translation framework, to try and unlock and accelerate innovation, bridging the fundamental breakthroughs with industrial translation. Then, through research results from ongoing work on drug delivery for cancer therapy, we will show how medical technology development benefits from manufacturing research. We will look then at how manufacturing research tackles larger themes, such as how to shift towards medical technology circularity, reducing waste and improving resilience. These different aspects of manufacturing research are being brought together in Cambridge in a new Institute for Biomedical Innovation and here we will describe the approach of considering the broader value chain when supporting end-to-end innovation.

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PLENARY SPEAKER

Prof. Yusuf Altintas
University of British Columbia, Canada

Biography: Professor Altintas is the coordinator Mechatronics Engineering and the director of Manufacturing Automation Laboratory at the University of British Columbia. He conducts research on metal cutting, machine tool vibrations, control, sensors and actuators for machine tools, and virtual machining. He has published 235 archival journal with over ~47,000 citations with an h index of 112, and a textbook: He is a Fellow of National Academy, Royal Society of Canada, CIRP, ASME, SME, ACATECH and Chief Editor of CIRP Journal and past president of CIRP. His research laboratory created advanced machining process simulation (CUTPRO), virtual part machining process simulation (MACHPRO) and a Virtual CNC, which are used by over 350 companies and research centers in the field of machining and machine tools worldwide.

Plenary Session: Mathematical Modelling of Machining Processes and CNC Machine Tools for Digital Twin Assisted Process Planning and Monitoring

Abstract: The presentation will provide an overview of the fundamental research conducted in our laboratory and its industrial applications as digital machining technology. The aim of our research is to develop mathematical models of metal cutting operations, machine tool vibrations, and control systems. These science-based digital models enable the virtual design of machine tools, as well as the simulation, optimization, and online monitoring of machining operations. The models predict cutting forces, torque, and power consumption during machining by considering material properties, cutter geometry, structural flexibility, and cutting conditions along the tool path. The dynamics of servo drive control systems and trajectory generation of machine tools are also considered. An in-house-developed virtual and real-time CNC system enables the design and analysis of any five-axis machine tool controller. The virtual machining system simulates the process and adjusts the feed rate along the tool path within a CAM environment. The system is also used for online monitoring and control of machine tools and machining processes by communicating with the CNC in real time as a digital twin.

PLENARY SPEAKER

Prof. Konstantinos (Kostas) Salonitis
University College Dublin, Ireland

Biography: Professor Konstantinos (Kostas) Salonitis is a Full Professor of Advanced Manufacturing and Director of the Research Ireland Centre on Advanced Manufacturing (I-Form) at University College Dublin since April 2026. He previously served as Head of the Sustainable Manufacturing Systems Centre and Professor of Manufacturing Systems at Cranfield University (2019–2026), joining Cranfield in 2012 after over a decade at the University of Patras, Greece. His research focuses on sustainable and resource-efficient manufacturing, process modelling and simulation, lean and green manufacturing, and environmental impact assessment. He has secured over £15 million in research funding, authored 350+ publications, and written two books. He is a Chartered Engineer and Fellow of multiple professional institutions (IMechE, IET, HEA).

Plenary Session: Sustainability and Additive Manufacturing

Abstract: Additive manufacturing (AM) is a transformative production approach with strong potential to support sustainability across the product lifecycle. Its layer-by-layer fabrication enables material efficiency, design flexibility, and decentralised production, helping reduce waste, energy use, and supply-chain complexity. This paper examines AM's role in environmental, economic, and social sustainability, focusing on material utilisation, lightweight design, energy performance, and circular economy strategies. Key enablers such as AM design, material recyclability, and on- demand production are discussed, alongside challenges including high energy intensity, feedstock sourcing, and end-of-life issues. Drawing on recent research and industrial applications, the paper identifies best practices and remaining gaps. The findings show that AM is not inherently sustainable; however, when combined with sustainable design principles and lifecycle assessment, it can significantly improve manufacturing sustainability. The paper concludes with future research directions and policy considerations to support responsible adoption of AM.

KEYNOTE SPEAKER

Prof. Ahmed Qureshi
University of Alberta, Canada

Biography: Dr. Ahmed Jawad Qureshi, Ph.D., P.Eng., is a professor and Research Chair in Advanced Manufacturing Processes and Automation at the University of Alberta, where he also serves as the founder and director of the Additive Design and Manufacturing Systems (ADaMS) Lab having a team of about 25 researchers and students. With over two decades of experience spanning North America, Europe, and Asia, Dr. Qureshi's research focuses on robotic hybrid additive-subtractive manufacturing, digital twinning, AI-driven design methodologies, and advanced material processing. His pioneering contributions to robotic additive manufacturing include developing hybrid manufacturing systems and advanced material processing for sectors such as aerospace, mining, energy, and healthcare. He has received over $20 million in funding from prestigious organizations, including the Natural Sciences and Engineering Research Council of Canada (NSERC), MITACS, and international government and industry partnerships. As an educator, mentor, and co- founder and CTO of Elementiam Materials and Manufacturing Inc., Dr. Qureshi bridges academic innovation with industrial impact, making him a visionary leader in advanced manufacturing and Industry 4.0 technologies.

Keynote Lecture: Including the Human Touch in Robotic Manufacturing and Industry 5 through Surrogate Process Digital Twinning

Abstract: This talk explores how the human touch can be captured and embedded into robotic manufacturing through surrogate process digital twinning for Industry 5.0. It highlights the need to develop sensor-fusion systems that combine force, vibration, and video-based motion analysis to understand how skilled operators perform complex manual manufacturing tasks. By collecting synchronized and annotated process data, these systems can translate human decision-making, dexterity, and process intuition into teachable robotic knowledge. The talk will discuss how such human-informed digital twins can accelerate robot training, improve adaptability in variable manufacturing environments, and support more collaborative, intelligent, and human-centered advanced manufacturing systems.

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KEYNOTE SPEAKER

Prof. Mozafar Saadat
University of Birmingham, UK

Biography: Dr Mozafar Saadat is with the Department of Mechanical Engineering, School of Engineering at the University of Birmingham, UK. He holds an Honours degree in Mechanical Engineering from University of Surrey, and a PhD in Industrial Automation from University of Durham, UK. His research interests are in automation, robotics, and intelligent manufacturing with research groups in autonomous remanufacturing and medical robotics including robotic rehabilitation and assistive reproductive technology. His innovations have led to the creation of a funded startup and three patents in infertility treatment. He has published widely in prestigious journals, supervised more than 30 PhD students, and obtained various research fundings from EU, EPSRC, Innovate UK, and industry.

Keynote Lecture: Manufacturing of Microfluidics-based Lab-on-a-Chip Medical Device for Cell Micromanipulation

Abstract: Infertility affects approximately 15% of couples worldwide. Intracytoplasmic sperm injection (ICSI) has become one of the most effective solutions in IVF clinics, where a single sperm is manually injected into an egg in a laboratory for subsequent embryo formation before being transferred to the patient. It involves manually selecting a healthy “sperm” from a small semen sample; a process that is highly dependent on the visual assessment and expertise of the embryologist. Microfluidics technology offers distinct advantages over the manual sperm selection, where the whole semen sample can be used in a high-speed, data-driven automated operation. It involves the miniaturisation of conventional fluidic processes into micron-scale devices, enabling reduced reagent usage and highly controlled fluid manipulation. However, the fabrication of such devices, especially those incorporating 2.5D structures or multiple length scales, can be expensive and time-consuming. This presentation outlines the application of microfluidics in cell manipulation as a medical device and reviews a number of current and modern manufacturing techniques.

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KEYNOTE SPEAKER

Prof. Fethi Abbasi

American University of the Middle East, Kuwait

Biography: Professor Fethi Abbassi received his Ph.D. in Mechanical Engineering and Mechanics of Materials from Toulouse University, France, in 2008. He is currently the Chair of Mechanical Engineering at the American University of the Middle East, Kuwait. Previously, he served as Assistant Dean, Department Chair and Associate Professor at Dhofar University, Oman, and as Assistant Professor at the University of Carthage, Tunisia. His research focuses on composite materials, nanocomposites, advanced manufacturing, 3D/4D printing, damage mechanics, and solid mechanics. He has published extensively in leading journals, with highly cited work and frequent keynote and plenary invitations. His contributions have gained international recognition, including a top-cited article in Engineering Results journal.

Keynote Lecture: Composites Materials: From Conventional to Advanced Manufacturing for Emerging Applications

Abstract: The integration of composite materials in modern engineering is highly demanded due to their high specific strength, design flexibility, and multifunctionality. This keynote examines their evolution from conventional fabrication to modern technologies for emerging applications. The talk begins with fiber-reinforced composites: synthesis, measurements and numerical modelling, enabling process optimization and structural integrity. The second part is a showcase of carbon-fibre reinforced polymer composites through experimental and numerical approaches to predict stiffness, strength, and damage under varied loading conditions. Advanced characterization tools such as Digital Image Correlation, acoustic emission, and µCT scanning are used to underpin the mechanism of crack-propagation-to-failure. Sustainability is addressed through natural fiber composites whose bio-inspired architecture demonstrates promising impact resistance and energy absorption capabilities. The role of interfaces in composite assemblies is also explored, focusing on adhesion, degradation and failure mechanisms to enhance life cycle and durability. Finally, advances in additive manufacturing of polymeric composites are presented, highlighting improved performance through integrated characterization and multiscale modelling approaches.

KEYNOTE SPEAKER

Dr. Ahmed Maged
American University of Sharjah, UAE

Biography: Dr. Ahmed Maged is an Assistant Professor in Industrial Engineering at the American University of Sharjah. His work focuses on applying machine learning and data-driven methods to intelligent manufacturing systems, with an emphasis on quality engineering, process monitoring, and anomaly detection.

He received his PhD in Systems Engineering from the City University of Hong Kong. His research centers on developing intelligent models for fault detection, defect detection, and predictive monitoring in complex industrial processes. His work addresses challenges such as high-dimensional data, nonlinear behavior, and time-dependent dynamics commonly found in modern manufacturing environments.

Dr. Maged has published in leading journals including IEEE Transactions on Industrial Informatics, IISE Transactions, and Applied Energy. His recent work explores representation learning, self-supervised learning, and uncertainty-aware models for improving reliability and decision-making in production systems.

Keynote Lecture: Intelligent Anomaly Detection in Manufacturing Systems Using Data - Driven Models

Modern manufacturing systems produce large amounts of process, sensor, and inspection data. Yet timely detection of anomalies, faults, and quality defects remains a major issue. Classical monitoring methods often depend on assumptions that are too strict for real production settings, especially when the data are high-dimensional, nonlinear, and time-dependent. This can lead to missed faults, delayed intervention, or excessive false alarms. This talk presents a data-driven view of anomaly detection for manufacturing systems, where machine learning models learn system behavior directly from data. The focus is on methods that combine representation learning, temporal modeling, and uncertainty-aware decision making to improve detection accuracy and monitoring stability. These methods are designed to better handle variation in operating conditions, complex process behavior, and changing production environments. Using case studies from manufacturing process monitoring and automated quality inspection, the talk shows how data-driven models can support earlier fault detection, reduce false alarms, and improve quality control decisions. The overall aim is to show how machine learning can support more reliable and practical monitoring in modern manufacturing.

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Conference Information:

Time Zone	GMT+05:00
Conference Date	September, 2-4, 2026
Link	https://us06web.zoom.us/j/3930618297?pwd=QVhNZlJCT0NlK2daekpPNnNyZjRDdz09

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ICMR 2026 Program Overview 2–4 September 2026	
Day 1: Wednesday, 2 September 2026	
08:30–09:15	**Registration (C3 - Red Hall)**
09:15–09:45	**Welcome and Formal Opening of the Conference: Professor Waqar Ahmad, President Professor Rehan Sadiq, Provost Professor Bjorn Birgisson, Vice-Provost for Research and Innovation Professor Essam Shehab, Acting Dean and General Chair of ICMR 2026 Conference Photograph – All delegates (C3 – Red Hall)**
09:45–11:00	**Industry-Academia Panel Discussion (C3 – Red Hall) "Made in Kazakhstan: Can We Build a Manufacturing Nation?" Panelists: Mr Pierre Lutz (Alstom), Mr Aksel Schmidt (Air Liquide), Mr Shynggys Mukhamedkali (Wabtec), Prof. Bjorn Birgisson (Nazarbayev University), Prof. Yusuf Altintas (University of British Columbia, Canada) Moderator: Prof. Essam Shehab (Nazarbayev University)**

11:00–11:30	**Coffee Break (C3 – Ground Floor)**

11:30-12:30	**Plenary Session 1: (C3 – Red Hall) Professor Ronan Daly, Institute for Manufacturing, Cambridge University, UK "Exploring the Role of Academic Manufacturing Research in Delivering New Medical Technologies" Chair: Prof. Huck Ng**
12:30-13:30	**Lunch (C2 - Ballroom Left)**
13:30-14.30	**Plenary Session 2: (C3 – Red Hall) Workshop: Smart Manufacturing Prof. Yusuf Altintas (University of British Columbia, Canada) Chair: Prof. Md. Hazrat Ali**

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Parallel Session 1 (C2 - Ballroom Right)				
Time	**Session 1A: Sustainable Systems in Manufacturing - 1 (Venue: C2 - Ballroom Right - A) Chair: Prof Kostas Salonitis**	**Session 1B: Manufacturing Ergonomics and Education (Venue: C2-Ballroom Right - B) Chair: Prof. Gulnur Kalimuldina**	**Session 1C: Asset Monitoring and Supply Chain (Venue: C2-Ballroom Right - C) Chair: Prof. Ahmed Maged**	**Session 1D: Additive and Hybrid Manufacturing - 1 (Venue: C2-Ballroom Right - D) Chair: Prof. Asma Perveen**
14:30-14:50	**Paper 8: Visual Transparency Signals in Sustainable Manufacturing: A Qualitative Analysis of Sustainability Imagery** *Authors: Balaussa Shaimakhanova Azubayeva and Andrew Thomas*	**Paper 20: Self-Powered Soft Robotic Exo-Glove with Integrated Triboelectric Tactile Sensors for Assistive Hand Function in Impaired Persons** *Authors: Bekzat Azamatov, Ali Oryn, Shabnam Yavary and Gulnur Kalimuldina*	**Paper 10: Design and Impact Mitigation Performance of Foam-Reinforced Multiwall Auxetic Tubular Structures** *Authors: Benrose Prasad, Fethi Abbassi and Sherif Araby*	**Paper 9: Comparative Study of Powder Feeding Behavior for Different Materials in Powder-Based Directed Energy Deposition (DED)** *Authors: Yeldar Otynshiyev, Alikhan Kalmakhanbet, Yi Nie, Christos Spitas and Boris Golman*
14:50-15:10	**Paper 70: Integrated Biotechnological Manufacturing of Steviol Glycosides** *Authors: Nurgul Nugumanova*	**Paper 63: Safe and Sustainable by Design for Operator 5.0 Decision-Support: A Sociotechnical System Perspective Translated from Safety-Critical Healthcare** *Authors: Firda Rahmadani and Mecit Can Emre Simsekler*	**Paper 81: Risk-Driven Dynamic Supply Chain Reconfiguration in Industry 4.0: An Empirical SEM Analysis [Online]** *Authors: Dasunika Ubeywarna and Sameh Saad.*	**Paper 39: Influence of vacuum on the mechanical properties and surface roughness of 3d printing assisted investment casting** *Authors: Wajid Hussain, Muslim Mukhtarkhanov, Md. Hazrat Ali and Essam Shehab*
15:10-15:30	**Paper 79: Towards a Staged Assessment Framework for Smart Monitoring in Sustainable Manufacturing** *Authors: Ziyad Sherif, David Butler and Shoaib Sarfraz*	**Paper 66: A Patient-Centered Digital Service-Production Framework for Telemedicine in Mental Health Settings** *Authors: Firda Rahmadani, Ahmed Al Blooshi, Maha Al Mufti, Alaa Alqaryuti, Nadeen Faraj, Mecit Can Emre Simsekler and Khalifa Al Meqbaali*	**Paper 47: Towards an integrated generative AI knowledge management system for energy-sector maintenance operations** *Authors: Ali Alsayegh and Tariq Masood*	**Paper 45: Cybersecurity in Additive Manufacturing: A Survey of Attack Vectors, Defense Mechanisms, and Future Directions [Online]** *Authors: Muhammad Abdullah Niazi, Essam Shehab and Md.Hazrat Ali*

Parallel Sessions 2 (Ballroom Right)				
Time	**Session 2A: Advanced Manufacturing Technologies (Venue: C2-Ballroom Right - A) Chair: Prof. Mozafar Saadat**	**Session 2B: Applications of Industry 4.0 (Venue: C2-Ballroom Right - B) Chair: Prof. Yerkin Abdildin**	**Session 2C: Applications of Machine Vision (Venue: C2-Ballroom Right - C) Chair: Prof. David Butler**	**Session 2D: Modelling and Simulation (Venue: C2-Ballroom Right - D) Chair: Prof. Amgad Salama**
15:30-15:50	**Paper 59: Manufacturability-Driven Redesign of a Wire Flame Spray Nozzle: Integrating Design for Manufacture and Computational Fluid Dynamics** *Authors: Muhammed Muneem, Mozafar Saadat and Majid Malboubi*	**Paper 3: Development of an Integrated Industrial IoT System for Real-Time Data Monitoring and Control** *Authors: Zhalgas Bolatbayev*	**Paper 12: Transformer-based Detectors (DETRs) for the Industrial Personal Protective Equipment (PPE) Detection** *Authors: Fadhlan Hafizhelmi Kamaru Zaman, Syahrul Afzal Che Abdullah, Kok Mun Ng and Mohd Farid Md Salleh*	**Paper 2: Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation [Online]** *Authors: Ahmed Basager and Abdullah Alrabghi*
15:50-16:10	**Paper 67: Leveraging Design for Manufacturing and Assembly and 3D Printing in Product Redesign** *Authors: Elia Abdulkarim, Mozafar Saadat and Majid Malboubi*	**Paper 80: DFMA and 3D Printing in Consumer Product Redesign: A Hair Dryer Case Study** *Authors: Elia Abdulkarim, Mozafar Saadat and Majid Malboubi*	**Paper 24: Automated inspection of returned MedTech devices using deep learning for computer vision [Online]** *Authors: Divya Tiwari, Ioan Alexandru Herdea, Matthew Mannix, Yuri Mejia, Fiona Charnley and Ashutosh Tiwari*	**Paper 53: Study of the influence of operating factors on the thermal operating mode of internal combustion engines** *Authors: Aidarali Tulenov, Yermek Baubekov, Baurzhan Shoibekov, Gabit Bakyt and Gabit Bekbolatov*
16:10-16:30	**Paper 68: Integrating Design for Manufacturing and Assembly with 3D Printing for the Optimization of a Wireless Headphone Assembly [Online]** *Authors: Rana Meligy, Mozafar Saadat and Majid Malboubi*	**Paper 31: A Conceptual Simulation-based Digital Twin Framework for Predictive Analysis and Optimisation of Operational Efficiency and Workforce Productivity [Online]** *Authors: Ghayda Diabat, Ian Pye, Rasoul Khandan, James Gao, Mahrukh Muzaffar, Van Dang Le, Chi An Le, Nguyen Danh Nguyen, Jamaluddin Mahmud and Chi Hieu Le.*	**Paper 78: Real-time dimensional measurement of hot forged components: a pixel-level analysis** *Authors: Muthair Hafeez, Muftooh Siddiqi and David Butler*	**Paper 50: Strategic Governance of Decentralised Autonomous Supply Chains: Macro-Environmental and Contextual Frictions [Online]** *Authors: Funlade Sunmola, Patrick Burgess, Memunat Lami Salihu-Yusuf and Hakeem Omolade Sunmola.*

16:30-17:00	**Coffee Break & Networking (Upper Atrium)**

Parallel Sessions 3 (Ballroom Right)				
Time	**Session 3A: Remanufacturing - 1 (Venue: C2-Ballroom Right - A) Chair: Prof. Mozafar Saadat**	**Session 3B: Advanced Manufacturing Systems (Venue: C2-Ballroom Right - B) Chair: Prof Kostas Salonitis**	**Session 3C: Smart Manufacturing (Venue: C2-Ballroom Right - C) Chair: Prof. Didier Talamona**	**Session 3D: Remanufacturing - 2 (Venue: C2-Ballroom Right - D) Chair: Prof. Lee Than**
17:00-17:20	**Paper 25: Recycled Carbon Fiber in Circular Economy Applications: Recent Advances and Developments in Sustainable Composites** **[Online]** *Authors: Faraj Alsubaie and Liu Yang*	**Paper 21: Marjan-R1: Design, Fabrication, Thrust Allocation, and Initial Hydrodynamic Parameterization of a Custom Fully-Actuated ROV** **[Online]** *Authors: Md Tarique Bin Hamid, Sarvat M. Ahmad, Masroor Khan, Syed Imtiaz Ali Shah and Ali Albeladi*	**Keynote Presentation 1** **Title: “Including the Human Touch in Robotic Manufacturing and Industry 5 through Surrogate Process Digital Twinning”** ***Speaker: Professor Ahmed Qureshi, University of Alberta, Canada.***	**Paper 55: Development of a hybrid oil heating system in conditions of a switched off diesel engine of a shunting diesel locomotive and analysis of its effectiveness** *Authors: Gabit Bakyt, Malik Zhamankulov, Kuantkhan Sarsenov, Berikkhan Tleuzhanov, Irina Ashirbayeva and Aizhan Makhanova*
17:20-17:40	**Paper 48: Agentic Generative AI for Aviation MRO Scheduling: A Capability-Gated Autonomy Framework** **[Online]** *Authors: Maryam Busakhar and Ali Alsayegh*	**Paper 43: A.C.T.I.V.E. - advanced cybernetic technology for intervention, versatility, and exploration: a rocker-bogie mecanum rover with Kalman-fused localisation, Bézier-curve navigation, and Limelight vision tracking** *Authors: Targyn Faizulla and Rustam Askaruly*		**Paper 60: Manufacturing Capability Maturity Pathways: A Comparative Study of Emerging Industrial Clusters in Kyrgyzstan, Malaysia and Indonesia** *Authors: Lee Lee Than*

17:40-18:00	**Paper 75: Feasibility of flax/PLA thermoplastic composite for sustainable wind turbine blade repair** *Authors: Edward Cant*	**Paper 72: Design and Integration of a Crank-Angle-Synchronised In-Cylinder Gas Sampling System for Heavy-Duty Diesel Engine Combustion Diagnostics** *Authors: Hameed Alshammari [Online]*	**Paper 74: Development of a Low-Cost Intelligent Robotic Waste Sorting System Based on Computer Vision and Autonomous Manipulation** *Authors: Dauren Baltabay*	**Paper 4: Characterization and Modeling of Non-Spherical Granular Materials for High-Temperature Thermal Energy Storage** *Authors: Aidana Boribayeva, Luis R. Rojas-Solórzano, Jennifer Curtis, Nicolin Govender and Boris Golman*

18:00–18:30	**NU Laboratory Visit (C4)**

18:30–20:00	**Conference Dinner (Ballroom Left)**

	Day 2: Thursday, 3 September 2026
09:00–09:30	**Registration**
09:30–10:30	**Plenary Session 3: (C3 – Red Hall)** **Professor Yusuf Altintas, University of British Columbia, Canada** **“Mathematical Modelling of Machining Processes and CNC Machine Tools for Digital Twin Assisted Process Planning and Monitoring”** **Chair: Prof. Asma Parveen**

10:30-11:30	**Coffee Break (C3 – Ground Floor)**

11.30-12:30	**Plenary Session 4: (C3 – Red Hall)** **Professor Kostas Salonitis, University College Dublin, Ireland** **“Sustainability and Additive Manufacturing”** **Chair: Prof. Essam Shehab**

12:30-14:00	**Lunch Break (C2 - Ballroom Left)**

Parallel Sessions 4 (C2 - Ballroom Right)

Time	Session 4A: Digital Modelling and Monitoring (Venue: C2-Ballroom Right - A) Chair: Prof. David Butler	Session 4B: Supply Chain for Manufacturing (Venue: C2-Ballroom Right - B) Chair: Prof Dmitry Sizdov	Session 4C: Smart Systems (Venue: C2-Ballroom Right - C) Chair: Prof. Sherif Gouda	Session 4D: Digital Modelling and Monitoring (Venue: C2-Ballroom Right - D) Chair: Prof. Mozafar Saadat
14:00-14:20	**Paper 32: Real-Time Localized Digital Twin for Quadrotor UAV Flight Monitoring** *Authors: Azim Dukenderov, Meirzhan Mazhenov, Alikhan Duisenbay, Md. Hazrat Ali and Essam Shehab*	**Paper 17: Effervescent Tablet-Based Nanofluid with Enhanced Thermal Characteristics and Neutron-Absorbing for Rapid Nuclear Reactors LOCA Response [Online]** *Authors: Naser Ali and Fatima Safar*	**Keynote Presentation 2** **Title: Intelligent Anomaly Detection in Manufacturing Systems Using Data-Driven Models** ***Speaker: Dr. Ahmed Maged*** ***American University Sharjah –UAE***	**Paper 40: CommandNet for Target Pursuit: Learning-Based Control Evaluation [Online]** *Authors: Almat Abdimalik, Nurtay Kaziyev and Ilyas Muhammad*
14:20-14:40	**Paper 38: Allowance-Constrained CAD-to-Point-Cloud Registration for Robotic Machining-Margin Analysis of Aircraft Panel Blanks** *Authors: Yan Yang, Zhongwei Li, Kai Zhong and Shi Yusheng*	**Paper 27: Methane Concentration Forecasting in Underground Mines Using Sensor Data: A Systematic Review of Methods, Datasets, and Open Challenges [Online]** *Authors: Iskander Akhmetov and Bakyt Kurmanov*		**Paper 54: Urban Heat Island Dynamics Around Industrial Zones Using Landsat and MODIS Time Series** *Authors: Shamma Albedwawi and Naeema Al Hosani*
14:40-15:00	**Paper 57: Development of an Intelligent Control System for an Integrated Low-Carbon Energy System Based on FPGA, ESP32, MQTT Protocol, and a Digital Twin** *Authors: Zhazira Shermantayeva*	**Paper 1: Agile methodologies in procurement contracts for improving new product development in complex engineering systems. [Online]** *Authors: Ayaulym Armanova*	**Paper 23: Self-Powered Triboelectric Electronic Skin with Braille-Based Interface for Assistive Communication in Deaf–Blind Individuals** *Authors: Raiymbek Nurtay, Olzhas Kanatbek, Shabnam Yavari and Gulnur Kalimuldina*	**Paper 56: The Decoupled Information Boundary: Resolving the Capacity Truth Paradox in Distributed Autonomous Shared Manufacturing Organisations [Online]** *Authors: Funlade Sunmola and George Baryannis*

Parallel Session 5 (C2 - Ballroom Right)				
Time	**Session 5A: Artificial Intelligence and ML in Manufacturing (Venue: C2-Ballroom Right - A) Chair: Prof. Didier Talamona**	**Session 5B: Manufacturing Optimisation (Venue: C2-Ballroom Right - B) Chair: Prof. Altay Zhakatayev**	**Session 5C: Sustainable Systems in Manufacturing - 2 (Venue: C2-Ballroom Right - C) Chair: Prof. Kostas Salonitis**	**Session 5D: Artificial Intelligence and ML in Manufacturing (Venue: C2-Ballroom Right - D) Chair: Prof. Fadhlan Hafizhelmi Kamaru Zaman**
15:00-15:20	**Paper 30: Deep Learning-Based Segmentation and Morphological Characterization of Gamma-Prime Precipitates in Superalloy Microstructures [Online]** *Authors: Isaac Iwediba and Vassili Vorontsov*	**Paper 11: The Application of Process Mapping and Simulation for Optimising Productivity in Low-Volume Assembly Cells [Online]** *Authors: Daniel Eaves, Andrew Rees, Andrew Thomas, Christian Griffiths and Balaussa Shaimakhan Azubayeva*	**Keynote Presentation 3** **Title: Manufacturing of microfluidics-based lab-on-a-chip medical device for cell micromanipulation** ***Speaker: Professor Mozafar Saadat, University of Birmingham, UK***	**Paper 19: From Prompts to Performance Rankings: Evaluating Large Language Models as Surrogates for Building Performance Simulation to Rank Residential Designs in Hot Climates** *Authors: Abdalrahman Alajmi and Ali Alsayegh*
15:20-15:40	**Paper 52: A Physics-Informed Machine Learning Framework for Defect Prediction in Directed Energy Deposition Using Simulation, Experimental Validation, and Generative AI [Online]** *Authors: Yerlik Gabdulla*	**Paper 71: A geometric-kinematic feasibility framework for robotic cable clip installation [Online]** *Authors: Yi Wang and Mozafar Saadat*		**Paper 37: Knowledge management in the age of generative AI: opportunities and challenges for Kuwait's energy sector** *Authors: Abdalrahman Alajmi and Ali Alsayegh*
15:40-16:00	**Paper 64: AI-driven voice biomarkers for early detection and continuous monitoring of schizophrenia** *Authors: Raghad Al-Sakaji, Firda Rahmadani, Haya Al Jaghoub, Shayma Alshehhi, Sarah Alkindi, Faisal Nawaz and Mecit Can Emre Simsekler*	**Paper 2: Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation [Online]** *Authors: Ahmed Basager and Abdullah Alrabghi*		**Paper 65: From Algorithmic Performance to Clinical Adoption: A Review-Based Taxonomy of Artificial Intelligence Clinical Decision Support Deployment Challenges** *Authors: Ahmed Saad, Firda Rahmadani, Khaled Toffaha, Mecit Can Emre Simsekler and Khalid Alzarooni*

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16:00-16:30	Coffee Break (Upper Atrium)			
Parallel Sessions 6 (C2 - Ballroom Right)				
Time	**Session 6A: Machining Processes and Advanced Materials (Venue: C2-Ballroom Right - A) Chair: Prof Asma Perveen**	**Session 6B: Automation, Flexible Assembly, and Sensors (Venue: C2-Ballroom Right - B) Chair: Prof. Fadhlan Hafizhelmi Kamaru Zaman**	**Session 6C: Additive and Hybrid Manufacturing - 2 (Venue: C2-Ballroom Right - C) Chair: Prof. Yerbol Sarbassov**	**Session 6D: Machining Processes and Advanced Materials (Venue: C2-Ballroom Right - D) Chair: Prof. Sherif Gouda**
16:30-16:50	**Paper 13: A Finite Element Study of the Graphene Coated-Bamboo Fibres Reinforced Polyethylene Composites. [Online]** *Authors: Faraj Alsubaie*	**Paper 7: IoT-Enabled Predictive Maintenance for Industrial Automation Systems** *Authors: Alisher Yelseitov, Zhalgas Bolatbayev and Tohid Alizadeh*	**Paper 28: Rapid Liquid Printing in Clay Medium for Construction-scale Concrete 3D Printing: A Process Development** *Authors: Mukhammad Safa Akimbay, Bakbergen Temirzakuly, Essam Shehab and Md. Hazrat Ali*	Keynote Presentation 4 **Title: Composite Materials: From Conventional to Advanced Manufacturing for Emerging Applications** *Speaker: Professor Fethi Abbassi American University of the Middle East – Kuwait*
16:50-17:10	**Paper 51: Integrated Thermodynamic Modeling and Process Optimization for Microstructure Control in LPBF-Processed CoCrNi–CuZr–Ta Medium-Entropy Alloys (MEAs)** *Authors: Clifford Omonini*	**Paper 42: Sensor-based physical safety system for human-robot collaboration** *Authors: Bakbergen Yermekbayev, Kaisar Kozhay, Muhammad Abdullah Niazi, Md Hazrat Ali and Essam Shehab*	**Paper 29: Geometry-aware inverse process planning for non-planar extrusion additive manufacturing** *Authors: Kaicheng Ruan, Yiwei Weng and Yi Xiong*	
17:10-17:30	**Paper 49: Development of Sustainable Aerated Geopolymer Concrete Using Steel Industry By-products** *Authors: Kassym Talgatuly, Chang-Seon Shon, Nazerke Kabylbek, Aigerim Baizhigitova and Saken Sandybay*	**Paper 14: Optimization of production rate of WEDM using Taguchi Methodology** *Authors: Rashmi Arya and Hari Singh*	**Paper 44: A comparative study of energy-efficient WEDM strategies for better surface integrity for machining of hybrid composite** *Authors: Nilesh Kumar, Jatinder Kumar and Rajesh Khanna*	

18:00-21:00	**Conference Gala Dinner (Ticket Holders Only) - (C2 - Ballroom Left)** **Award Ceremony and Closing Remarks**

Day 3: Friday, 4 September 2026 - City Visit	
08:30-09:00	**Registration (Block 1 - Ground Floor)**
09:00-12:30	**City Visit**

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CONFERENCE ABSTRACTS

<table>
<tr><th colspan="2">Session 1A: Sustainable Systems in Manufacturing - 1
(Venue: C2 - Ballroom Right - A)
Chair: Prof Kostas Salonitis</th></tr>
<tr><th>Time</th><th>Content</th></tr>
<tr><td>14:30-14:50
Sep.2</td><td>Paper 8: Visual Transparency Signals in Sustainable Manufacturing: A Qualitative Analysis of Sustainability Imagery
Authors: Balaussa Shaimakhanova Azubayeva, Andrew Thomas
Abstract: Sustainability has become a central concern for manufacturing organisations, accompanied by increasing scrutiny of the credibility and transparency of sustainability communication. While existing research on sustainable manufacturing has largely focused on textual disclosures, metrics, and certifications, manufacturers increasingly rely on visual material—such as images of facilities, processes, products, and people—to support sustainability claims. Despite their widespread use, the role of such visuals in demand-side evaluation of manufacturing sustainability remains underexplored. Drawing on signalling theory, this study conceptualises sustainability-related manufacturing imagery as a form of visual transparency signalling used under conditions of information asymmetry. To address this exploratory and interpretive research objective, the paper adopts a systematic qualitative visual content analysis of sustainability-related imagery published by sustainability award-winning manufacturing organisations. Images are collected from publicly available sustainability reports, corporate websites, and award-related communications, and are analysed using a structured coding framework capturing operational processes, product stewardship, social responsibility, environmental symbolism, and governance cues. The study makes three contributions to manufacturing research. First, it extends signalling theory by identifying how visual representations of manufacturing processes function as sustainability and transparency signals. Second, it introduces a replicable analytical framework for examining sustainability-related manufacturing imagery, addressing a methodological gap in sustainability and transparency research. Third, by adopting a demand-side perspective, the study advances understanding of how manufacturing practices are evaluated for credibility and trust, highlighting implications for accountability and greenwashing in contemporary manufacturing systems. The findings offer practical guidance for manufacturers seeking to enhance transparency and for stakeholders assessing the credibility of sustainability claims.</td></tr>
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14:50-15:10 Sep.2	**Paper 70:** Integrated Biotechnological Manufacturing of Steviol Glycosides **Authors:** ***Nurgul Nugumanova*** **Abstract:** Demand for low-calorie, naturally occurring sweeteners has fueled interest in steviol glycosides, which accumulate in the leaves of Stevia rebaudiana Bertoni. Steviol glycosides have become a key alternative to sugar due to their high sweetening intensity, safety, and lack of addiction. However, production methods have not kept pace with demand for the purest, most flavorful minor components. This paper examines their production as a production challenge, and asks how the industry can transition from an unstable, agriculturally dependent supply to controlled, scalable, and sustainable industrial production. This work provides a brief overview and does not exhaust all the challenges encountered. Data from industrial biotechnology, bioprocess engineering, separation science, and digital manufacturing were summarized and reorganized around production line stages so that each technology could be assessed based on production criteria - productivity, product stability, downstream processing load, cost, and resource consumption. Traditional cultivation methods and solvent extraction, plant tissue culture, callus, suspension, and hairy root cultures, bioreactor cultivation, enzymatic conversion, and microbial fermentation were placed on a unified basis, along with the extraction and purification stages that ultimately determine product quality. Today, industrial production relies almost entirely on cultivated leaves, which are characterized by genetic instability, a heterogeneous glycoside profile, and low levels of valuable compounds. Extraction of these minor compounds is inefficient, and purification remains multi-step and expensive. Two ingenious methods, both outside of the plant itself, have been proposed to address this problem. Enzymatic conversion and microbial fermentation generate specific profiles of valuable glycosides independent of leaf composition and are already closest to industrial application, while plant cell and organ cultures remain primarily at the laboratory level. "Greener," enhanced extraction—ultrasonic, microwave, and pressurized water - along with membrane clarification can shorten processing lines and reduce solvent and energy consumption. No single method is suitable for all products: bulk glycosides are suited to cultivation and improved biomass, while rare ones gravitate towards fermentation and biocatalysis, so production is likely to become hybrid rather than universal. A hybrid approach is most effective when considered as parts of a single system. We propose an integrated biotech production platform (Fig. 1) in which plant biomass, fermentation, and enzymatic processing feed a single, intensified processing line, while quality control, artificial intelligence (AI)-based process optimization, and process analytical technology (PAT) operate across all stages. Advances in machine learning and digital twin modeling make such coordination increasingly practical, enabling prediction and adjustment of processes before implementation and maintaining quality control in real time. This approach distinguishes this review: rather than focusing on the biology of stevia or a single processing method, it links biotechnology, production engineering, downstream processing, sustainability, and digital biomanufacturing into a single industrial picture and uses it to identify where value and risk truly lie along the production chain. Commercial adoption of these new methods is still limited, with few feasibility studies and life cycle assessments. Validation of the engineering and enzymatic processes at scale is necessary before the hybrid platform can be considered proven. Regulatory approval and consumer perception of fermentation-based sweeteners will also

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	influence adoption. Nevertheless, the overall trend is stable: steviol glycoside production is shifting from an agricultural raw material model to a digitally coordinated, sustainability-focused industrial biotechnology industry.
15:10-15:30 Sep.2	**Paper 79:** Towards a Staged Assessment Framework for Smart Monitoring in Sustainable Manufacturing **Authors:** ***Ziyad Sherif, David Butler, Shoaib Sarfraz*** **Abstract:** Smart monitoring is increasingly promoted as a route to more sustainable manufacturing, yet monitoring capability does not necessarily translate into operational decisions or verified improvement. This paper proposes a staged assessment framework that shifts evaluation from installed capability to decision-readiness before deployment and verified outcomes afterwards. The framework is anchored by a four-stage Data Usage Progression (DUP) lens (Collection, Reporting, Decision-Enabling and Outcome Measurement) and combines a pre-implementation readiness assessment, a post-implementation outcome-verification assessment, a decision register and a five-level maturity scale. A literature-informed precision-machining example illustrates how the framework structures monitoring requirements, decision ownership and evidence paths. The paper contributes an initial assessment logic for aligning smart monitoring with sustainable operational performance and identifies industrial validation and maturity-scale calibration as the next stages of development.

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Session 1B: Manufacturing Ergonomics and Education

(Venue: C2-Ballroom Right - B)

Chair: Prof. Gulnur Kalimuldina

Time	Content
14:30-14:50 Sep.2	**Paper 20:** Self-Powered Soft Robotic Exo-Glove with Integrated Triboelectric Tactile Sensors for Assistive Hand Function in Impaired Persons **Authors:** ***Bekzat Azamatov, Ali Oryn, Shabnam Yavary, Gulnur Kalimuldina*** **Abstract:** Hand impairments significantly limit the ability of individuals to perform routine daily activities such as grasping, holding, and manipulating objects. Although several assistive exoskeleton gloves have been proposed to address these challenges, many existing systems remain expensive, bulky, mechanically complex, and lack reliable tactile sensing for intuitive control. To address these limitations, this work presents a self-power soft robotic assistive exo-glove that integrates tendon-driven actuation with triboelectric nanogenerator (TENG)–based tactile sensing, enabling both mechanical assistance and real-time touch detection. The complete system was first designed using SolidWorks computer-aided design (CAD) to optimize the glove geometry, tendon routing, and actuator placement for efficient finger motion and ergonomic wearability (Fig 1a). Following the design stage, the glove structure was fabricated using a 3D silicone printing process with SIL-30 elastomer. This material was selected due to its high elasticity, mechanical durability, and ability to withstand repeated bending during finger movement (Fig 1b). The soft printed structure enables natural hand motion while maintaining sufficient mechanical support for assistive operation. For mechanical assistance, a compact tendon-driven actuation mechanism was developed. In this design, motor rotation is converted into linear displacement through a screw-guided transmission system, allowing controlled pulling and releasing of tendons routed along the glove fingers. This mechanism enables controlled finger flexion and extension while maintaining a lightweight and compact structure. To provide tactile sensing capability, a triboelectric nanogenerator (TENG)-based tactile sensor was integrated at the fingertips of the glove. The sensor was fabricated using a microprinting technique, allowing the formation of micro-scale sensing structures directly on flexible substrates. The sensing architecture consists of conductive silver nanoparticles electrodes sandwiched between two silicone layers, forming a flexible triboelectric interface capable of converting mechanical contact into electrical signals. The microfabricated design enables rapid response and high sensitivity to touch interactions. The electrical performance of the TENG tactile sensor was evaluated under both manual tapping and controlled mechanical loading conditions. During natural finger tapping, the sensor generated alternating voltage peaks exceeding 40 V, demonstrating strong responsiveness to human touch. Under controlled loading conditions, static forces ranging from 1–5 N and cyclic contact frequencies between 1–3 Hz were applied (Fig 1d). The generated electrical output closely followed the applied force magnitude and contact frequency, confirming

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	reliable sensing behavior for realistic touch interactions. To evaluate the mechanical suitability of the printed silicone material, tensile tests were performed on five ASTM D12 standard specimens using a universal tensile testing machine. The material exhibited an average maximum tensile strength of 2.08 MPa and an average elongation at break of 885%, indicating excellent elasticity and flexibility suitable for wearable soft robotic applications (Fig 1c). The fully assembled system combines soft robotic actuation with self-powered tactile sensing in a single wearable platform. Experimental results demonstrate that the tendon-driven mechanism enables effective finger flexion and extension, while the integrated TENG sensor provides reliable touch detection that can be used to control the motor-driven tendon system for opening and closing the fingers. Overall, the proposed self-power soft robotic exo-glove demonstrates a promising approach toward affordable, lightweight, and intelligent assistive devices for individuals with hand impairments. The integration of CAD-based design, 3D silicone printing, microfabricated triboelectric tactile sensors, and tendon-driven actuation provides a scalable platform for future wearable rehabilitation technologies and smart assistive systems.
14:50-15:10 Sep.2	**Paper 63:** Safe and Sustainable by Design for Operator 5.0 Decision-Support: A Sociotechnical System Perspective Translated from Safety-Critical Healthcare **Authors:** ***Firda Rahmadani and Mecit Can Emre Simsekler*** **Abstract:** As AI-based decision-support becomes embedded in human–machine collaborative assembly and maintenance, Industry 5.0 reframes the operator as the human-centric core of the production system rather than a cost to be automated away. Yet operator-facing decision-support is still commonly optimized for task accuracy and throughput, while the sociotechnical conditions that make it safe to rely on and sustainable to work alongside are addressed only after deployment problems surface. This paper extends the Safe and Sustainable by Design (SSbD) principles to the design of operator-facing AI decision support within the Operator 5.0 paradigm. We conceptualize it through a human-factors framework built on two pillars: safe-by-design (trust calibration, workload-aware interaction, meaningful human control, and resilient degradation) and sustainable-by-design (energy-efficient deployment, protection against deskilling and burnout, and organizational adoptability). Design principles are derived from evidence in safety-critical healthcare decision support and illustrate using an adaptive human–machine collaborative work.

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15:10-15:30 Sep.2	**Paper 66:** A Patient-Centered Digital Service-Production Framework for Telemedicine in Mental Health Settings **Authors:** ***Firda Rahmadani, Ahmed Al Blooshi, Maha Al Mufti, Alaa Alqaryuti, Nadeen Faraj, Mecit Can Emre Simsekler, Khalifa Al Meqbaali*** **Abstract:** The rising demand for mental health services highlights the need for accessible, scalable, and effective care delivery models. Telemedicine offers a transformative solution by leveraging technology to address gaps in access and quality. This paper reviews and assesses different tools and models used in mental health settings and proposes a patient-centered framework for implementing telemedicine in mental healthcare, incorporating the perspectives of key stakeholders: patients, healthcare providers, regulators, and insurers. The framework integrates the use of various technologies to support telemedicine in healthcare while highlighting the main challenges from different stakeholder perspectives. This framework lays a foundation for advancing telemedicine in mental health with recommendations for future research and policy development. This study uses a systems thinking approach to evaluate telemedicine tools and models, focusing on seamless care delivery, stakeholder collaboration, and patient-centric process design. A general framework for implementing telemedicine in mental healthcare that emphasizes personalized care pathways and iterative feedback loops to enhance patient outcomes.

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Session 1C: Asset Monitoring and Supply Chain (Venue: C2-Ballroom Right - C) Chair: Prof. Ahmed Maged	
Time	**Content**
14:30-14:50 Sep.2	**Paper 10:** Design and Impact Mitigation Performance of Foam-Reinforced Multiwall Auxetic Tubular Structures **Authors:** ***Benrose Prasad, Fethi Abbassi, Sherif Araby*** **Abstract:** Auxetic structures, characterized by a negative Poisson's ratio, have garnered considerable attention for impact mitigation applications due to their high energy absorption capacity and enhanced deformation stability. This study presents a comprehensive parametric numerical investigation into the mechanical performance of foam-filled concentric tubular structures reinforced with additively manufactured re-entrant honeycomb composites. The proposed hybrid system is conceived as a lightweight, high-performance energy absorber for advanced engineering applications. Multiwall tubular auxetic structures (MTAS) with various concentric configurations are developed and integrated as reinforcements within foam cores. Their mechanical response under axial impact loading is systematically evaluated using finite element simulations. Nonlinear deformation, progressive collapse mechanisms, and energy absorption behaviour are captured through detailed numerical modelling. Specific energy absorption, peak crushing force, and deformation analysis are used to quantitatively evaluate the performance of different geometric and material configurations. Results demonstrate that the synergistic coupling between auxetic multiwall tubes and foam significantly enhances energy absorption while maintaining stable crushing behaviour and reduced force transmission. The proposed auxetic tubular system exhibits strong potential for use in crash cushions design, marine impact protection systems, and aerospace and space structures, where lightweight and efficient energy absorbers are critical. The findings provide valuable design insights for emerging applications.

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14:50-15:10 Sep.2	**Paper 81:** Risk-Driven Dynamic Supply Chain Reconfiguration in Industry 4.0: An Empirical SEM Analysis **Authors:** ***Dasunika Ubeywarna, Sameh Saad*** **Abstract:** Global supply chains are increasingly exposed to disruptions arising from volatile markets, geopolitical uncertainty, and technological complexity, highlighting the need for more adaptive and resilient supply chain designs. Although Industry 4.0 technologies enhance visibility, connectivity, and decision-making capabilities, empirical evidence on their role in supporting risk-driven dynamic supply chain reconfiguration remains limited. This study investigates dynamic supply chain reconfiguration within digitally enabled manufacturing environments using a Structural Equation Modelling (SEM) approach implemented in IBM SPSS AMOS Version 26. The proposed framework integrates five key dimensions: internal risk, external risk, environmental risk, physical aspects, and business processes. Data were collected through a structured survey of academic and industry professionals. Reliability analysis confirmed acceptable internal consistency, while the final model demonstrated satisfactory fit (CFI = 0.946, TLI = 0.922, RMSEA = 0.077). The findings highlight the importance of aligning digital technologies, network design, and business process capabilities to support adaptive and resilient supply chain reconfiguration. The validated framework provides both theoretical insights and practical guidance for organisations seeking to enhance risk management and resilience in Industry 4.0 environments.
15:10-15:30 Sep.2	**Paper 47:** Towards an integrated generative AI knowledge management system for energy-sector maintenance operations **Authors:** ***Ali Alsayegh, Tariq Masood*** **Abstract:** Maintenance knowledge in critical energy infrastructure is fragmented across manuals, drawings, operational records and the tacit expertise of an ageing and often bilingual workforce. Generative artificial intelligence (GenAI) can support knowledge capture, retrieval and reuse, but many current applications remain isolated tools rather than components of an integrated knowledge management system. This limitation is particularly important in energy-sector maintenance, where knowledge is bilingual, equipment-specific and distributed across both documents and experienced personnel. Drawing on the literature and a corpus of operational documents collected from Kuwait's Ministry of Electricity and Water, this paper derives eight design recommendations for GenAI-enabled knowledge management systems in maintenance contexts. The recommendations are operationalised through a six-layer reference architecture, a closed governance workflow and an illustrative clickable prototype. The novelty lies in integrating bilingual retrieval, expert–equipment linkage, citation-grounded generation and expert-gated governance within one maintenance-specific architecture.

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<table>
<tr><th colspan="2">Session 1D: Additive and Hybrid Manufacturing - 1
(Venue: C2-Ballroom Right - D)
Chair: Prof. Asma Perveen</th></tr>
<tr><th>Time</th><th>Content</th></tr>
<tr><td>14:30-14:50
Sep.2</td><td>Paper 9: Comparative Study of Powder Feeding Behavior for Different Materials in Powder-Based Directed Energy Deposition (DED)
Authors: Yeldar Otynshiyev, Alikhan Kalmakhanbet, Yi Nie, Christos Spitas, Boris Golmans
Abstract: Powder-based Directed Energy Deposition (DED) is an additive manufacturing process in which metallic powder particles are transported by inert gases into a laser-induced melt pool, where they interact with the laser beam, melt, and solidify, fabricating metallic components. Inert gases play a critical role in the powder feeding process by preventing oxidation and ensuring stable powder transport. In powder-based DED systems, inert gases are commonly used in the form of carrier gas, shielding gas, and optical gas, among which the carrier gas is responsible for delivering powder particles into the melt zone. Powder feeding is a crucial initial stage of the DED process, as it directly influences particle–particle interactions, particle trajectories, and the spatial distribution of powder at the stand-off distance. Variations in gas flow characteristics and material properties can significantly affect powder stream stability and deposition behavior. This study investigates the powder feeding behavior of SS316L and CuSn10, examining the effects of carrier gas and shielding gas, as well as powder feeding parameters, on powder stream characteristics. Experimental investigations were conducted using a high-speed camera to analyze powder streamlines and particle flow velocities. The results show that the powder convergence angle increases with increasing shielding gas flow rate for both SS316L and CuSn10. SS316L exhibits a more pronounced widening of the powder stream compared to CuSn10, while variations in powder feeder rotational speed and carrier gas flow also influence the convergence angle, indicating material-dependent powder feeding behavior in powder-based DED.</td></tr>
<tr><td>14:50-15:10
Sep.2</td><td>Paper 39: Influence of vacuum on the mechanical properties and surface roughness of 3d printing assisted investment casting
Authors: Wajid Hussain, Muslim Mukhtarkhanov, Md. Hazrat Ali, Essam Shehab
Abstract: Rapid investment casting (RIC) is a great tool for manufacturing highly detailed small metallic parts in low to medium volume. Thus, it found application in areas such as jewelry and dentistry. In most cases, to achieve high detalization, people resort to vacuum-assisted investment casting. Vacuuming during casting facilitates better flow of the molten metal and prevents premature solidification. A downside of this method is the high speed of metal flow, which can lead to</td></tr>
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	turbulence and air entrapment. This study analyzes the effect of vacuuming on the mechanical properties of investment-cast aluminum alloy by comparing the surface roughness and tensile strength of the specimens manufactured with and without vacuuming. The analysis of variance (ANOVA) confirms that vacuuming has a highly significant effect on average surface roughness ($p < 0.001$). However, the tensile strength shows no statistically significant difference ($p > 0.718$) between vacuum casting and without vacuum (gravity) casting, at 139 ± 6.42 MPa and 137 ± 6.42 MPa, respectively.
15:10-15:30 Sep.2	**Paper 45:** Cybersecurity in Additive Manufacturing: A Survey of Attack Vectors, Defense Mechanisms, and Future Directions **Authors:** ***Muhammad Abdullah Niazi, Essam Shehab, Md.Hazrat Ali*** **Abstract:** A wide range of safety-critical domains, including healthcare, automotive, aerospace, and defense applications, have adopted Additive Manufacturing (AM) in production. However, AM workflows such as digital design, network communication, firmware, and physical fabrication introduce a cyber-physical attack surface that, if compromised at earlier stages, can result in fabrication-level defects. This paper covers cybersecurity challenges in AM and presents a taxonomy consisting of four layers: design, network, firmware, and physical fabrication. The paper examines cyberattacks in AM systems, including malicious G-code attacks, network attacks, side-channel attacks, and firmware attacks. It also surveys existing mitigation methods, such as cryptographic protection, monitoring, physical security, and network security. Most existing approaches work in isolation at a single layer and do not provide complete end-to-end security coverage across the AM workflow. The paper identifies security flaws, including the lack of lightweight security mechanisms for resource-constrained devices and the need for attack-resistant architectures in AM.

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Session 2A: Advanced Manufacturing Technologies

(Venue: C2-Ballroom Right - A)

Chair: Prof. Mozafar Saadat

Time	Content
15:30-15:50 Sep.2	**Paper 59:** Manufacturability-Driven Redesign of a Wire Flame Spray Nozzle: Integrating Design for Manufacture and Computational Fluid Dynamics **Authors:** ***Muhammed Muneem, Mozafar Saadat and Majid Malboubi*** **Abstract:** This study applies Design for Manufacturing principles to redesign a MetalCoat 12M wire flame spray nozzle by replacing the original angled micro-hole geometry with a two-part annular-gap concept. The original nozzle and siphon-plug assembly were reverse engineered using optical metrology and reconstructed in Autodesk Fusion. Steady-state thermal analysis and cold-flow CFD were then used to compare the original and annular-gap geometries. Supporting thermal analysis showed that the annular-gap redesign also reduced peak simulated temperature by 1.8%, indicating that the simplified geometry did not compromise thermal behaviour. CFD results showed that the annular-gap design achieved a comparable maximum velocity with an increase of 0.3% to the original nozzle, while also producing a 21 % increase in velocity at the wire-exit region. A 3D printed prototype was additionally produced, validating the redesign's effectiveness and its capability for continuous flow. Overall, the results demonstrate that the annular-gap redesign improves manufacturability while preserving the key flow-convergence behaviour of the original nozzle.
15:50-16:10 Sep.2	**Paper 67:** Leveraging Design for Manufacturing and Assembly and 3D Printing in Product Redesign **Authors:** ***Elia Abdulkarim, Mozafar Saadat and Majid Malboubi*** **Abstract:** "This paper presents a Design for Manufacturing and Assembly (DFMA)–based redesign of a commercial Panasonic hair dryer, with a focus on component-level assembly inefficiencies. The product was disassembled reconstructed in CAD, and evaluated using the Lucas method and the Boothroyd–Dewhurst Design for Assembly (DFA) methodology. The initial DFA analysis identified the motor housing, front guard assembly, and hinge interface as the principal areas of concern, and the redesign therefore concentrated on these components. 3D printing was employed to manufacture the redesigned parts. A

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	cantilever snap-fit feature was developed for the motor housing and validated using finite element analysis. An asymmetrical placement of the snap-fit features was introduced to prevent motor rotation and reduce part count. The redesign reduced part count by 29.4%, the handling ratio by 31.6%, the fitting ratio by 34.5%, and the predicted assembly time from 120.57s to 87.04s. Overall, the results demonstrate that targeted DFMA modifications can deliver substantial assembly-time savings in high-volume consumer appliances while preserving manufacturability."
16:10-16:30 Sep.2	**Paper 68:** Integrating Design for Manufacturing and Assembly with 3D Printing for the Optimization of a Wireless Headphone Assembly **Authors:** ***Rana Meligy, Mozafar Saadat and Majid Malboubi*** **Abstract:** "This study presents a Design for Manufacture and Assembly redesign of the TaoTronics SoundSurge 90 wireless headphones to improve manufacturability and assembly efficiency while preserving functionality and geometry. The product was reverse engineered and reconstructed as a Computer-Aided Design model, from which a DFMA optimised redesign was developed. The redesign was evaluated using the Lucas Design for Assembly method and validated through finite element analysis and stereolithography prototyping. The redesign reduced the total number of components from 98 to 42 and fasteners from 40 to 2. Consequently, Design Efficiency increased from 33.67% to 76.19%. Prototype testing confirmed successful assembly and preservation of the original folding mechanism. The results demonstrate that DFMA can substantially simplify consumer compromising functionality."

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Session 2B: Applications of Industry 4.0 (Venue: C2-Ballroom Right - B) Chair: Prof. Yerkin Abdildin	
Time	**Content**
15:30-15:50 Sep.2	**Paper 3:** Development of an Integrated Industrial IoT System for Real-Time Data Monitoring and Control **Authors:** ***Zhalgas Bolatbayev*** **Abstract:** "The rapid growth of digitalization in the manufacturing sector has driven increased interest in Industrial Internet of Things (IIoT) technologies as a key enabler of digital twin implementations. Among the various components of digital twins, reliable real-time data acquisition from industrial equipment remains a critical challenge. This paper focuses on the development of an integrated IIoT framework for collecting, monitoring, and storing operational data from programmable logic controllers (PLCs). The proposed methodology involves acquiring real-time data from PLC tag values configured and managed using Siemens TIA Portal. Communication between the PLC and the IIoT system is achieved through the S7 protocol, implemented via Node-RED, which acts as a middleware layer for data acquisition, processing, and routing. The collected data is stored in multiple database systems, including time-series databases and cloud-based storage solutions such as Google Sheets and Google Cloud services, enabling scalable data analysis and visualization. The developed system provides a flexible and extensible architecture for integrating shop-floor automation systems with cloud-based data platforms. This approach supports real-time visualization and establishes a foundation for future data interpretation, advanced analytics, and decision-support applications in manufacturing environments."
15:50-16:10 Sep.2	**Paper 80:** DFMA and 3D Printing in Consumer Product Redesign: A Hair Dryer Case Study **Authors:** ***Elia Abdulkarim, Mozafar Saadat and Majid Malboubi*** **Abstract:** "This paper presents a Design for Manufacturing and Assembly (DFMA)–based redesign of a commercial Panasonic hair dryer, with a focus on component-level assembly inefficiencies. The product was disassembled, reconstructed in CAD, and evaluated using the Lucas method and the Boothroyd–Dewhurst Design for Assembly (DFA) methodology. The initial DFA analysis identified the motor housing, front guard assembly, and hinge interface as the principal areas of concern, and the redesign therefore concentrated on these components. 3D printing was employed to manufacture the redesigned parts. A cantilever snap-fit feature was developed for the motor housing and validated using finite element analysis (FEA). An asymmetrical placement of the snap-fit features was introduced to prevent motor rotation and reduce part count. The redesign reduced part count by 29.4%, the handling ratio by 31.6%, the fitting ratio by 34.5%, and the predicted assembly time from 120.57 s to 87.04 s. Overall, the results demonstrate that targeted DFMA modifications can deliver substantial assembly-

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	time savings in high-volume consumer appliances while preserving manufacturability."
16:10-16:30 Sep.2	**Paper 31:** A Conceptual Simulation-based Digital Twin Framework for Predictive Analysis and Optimisation of Operational Efficiency and Workforce Productivity [Online] **Authors:** ***Ghayda Diabat, Ian Pye, Rasoul Khandan, James Gao, Mahrukh Muzaffar, Van Dang Le, Chi An Le, Nguyen Danh Nguyen, Jamaluddin Mahmud and Chi Hieu Le.*** **Abstract:** "This study proposes a framework for developing a predictive and analytical simulation-based digital twin to optimise operational efficiency and workforce productivity, with a focus on modern high-mix and low-volume manufacturing industries, which face significant operational complexities due to product customisation, demand variability and resource constraints. A real-world case study was conducted for the predictive analysis of multiple-criteria scenarios at a test chamber facility, in which simulation serves as the essential bridge between the physical system and its virtual model, replicating and analysing the behaviour and performance of a test chamber facility within the digital twin environment. What-If scenario analyses were conducted to stress-test the system across three core variables: operator scaling, product failure rates, and peak demand. The framework establishes a practical and cost-effective solution for data-driven operational decision-making and pre-deployment planning in smart manufacturing, serving as a foundation for the full development of a digital twin platform integrating emerging AI agents and API-connected dashboards to further support data-driven and data-informed operational decisions."

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Session 2C: Applications of Machine Vision

(Venue: C2-Ballroom Right - C)

Chair: Prof. David Butler

Time	Content
15:30-15:50 Sep.2	**Paper 12:** Transformer-based Detectors (DETRs) for the Industrial Personal Protective Equipment (PPE) Detection **Authors:** ***Fadhlan Hafizhelmi Kamaru Zaman, Syahrul Afzal Che Abdullah, Kok Mun Ng and Mohd Farid Md Salleh*** **Abstract:** "Industrial and construction environments remain high-risk sectors where Personal Protective Equipment (PPE) serves as a critical final defense against workplace fatalities. While deep learning technologies like Convolutional Neural Networks (CNNs) and the YOLO series are commonly deployed for safety monitoring, they often struggle with the trade-off between real-time processing speeds and robust detection accuracy in complex, dynamic environments. This paper proposes a novel solution utilizing end-to-end Transformer-based detectors (DETRs), specifically the Real-Time Detection Transformer (RT-DETR) and its Neural Architecture Search-optimized variant, RF-DETR. Utilizing the diverse SH17 dataset for benchmarking, we evaluated multiple variants against state-of-the-art YOLOv11 and YOLOv26 models. Experimental results demonstrate that DETR-based architectures exhibit significant performance advantages, particularly in terms of convergence speed and stability. The RFDETRLarge model emerged as the superior architecture, achieving the highest mAP@50 score of 0.7399. Furthermore, RF-DETR variants demonstrated the ability to process images at 27–29 img/s on GPU, meeting the requirement for live safety monitoring. Our findings confirm that RF-DETR provides a more reliable foundation for mission-critical PPE detection by identifying human elements and safety gear with high spatial accuracy and minimal false alarms."
15:50-16:10 Sep.2	**Paper 24:** Automated inspection of returned MedTech devices using deep learning for computer vision [Online] **Authors:** ***Divya Tiwari, Ioan Alexandru Herdea, Matthew Mannix, Yuri Mejia, Fiona Charnley and Ashutosh Tiwari*** **Abstract:** "Communication remains a major challenge for individuals who simultaneously experience visual and hearing impairments. Most existing assistive communication systems rely on visual displays or auditory feedback, limiting their accessibility for deaf-blind users. To address this challenge, this study presents a self-powered electronic skin (e-skin) communication interface based on Braille language patterns, enabling tactile interaction through dot-based input while integrating triboelectric nanogenerator (TENG) sensing for signal generation and

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	recognition. The proposed e-skin was designed as a flexible tactile interface fabricated using silicone 3D printing technology. All structural components were first designed using SolidWorks CAD software and then fabricated using an S300 silicone 3D printer (Fig 1a). The sensing elements consist of triboelectric nanogenerators, where copper foil electrodes are embedded between two layers of printed silicone. To ensure stable bonding between silicone and copper electrodes, a two-component Ecoflex 00-30 elastomer was used during assembly. The final device structure was thermally cured to obtain a flexible, durable, and wearable electronic skin (Fig 1b, 1c). The upper silicone layer of the e-skin incorporates Braille-style tactile dots arranged in a 3 × 2 matrix, representing the Braille writing system used by visually impaired individuals (Fig 1b). By touching or pressing the raised Braille dots, users generate mechanical contact that activates the triboelectric effect between the silicone layers and copper electrodes. As a result, the system produces electrical signals without requiring an external power source (Fig 1d). This self-powered sensing capability enables the device to operate using the mechanical energy generated by finger interaction, making the interface suitable for wearable assistive communication. To interpret the tactile input patterns, a machine-learning-based signal recognition framework was developed. The device integrates an array of 16 independent TENG sensors, each corresponding to a specific spatial location on the Braille interface. Signals are acquired using a CD74HC4067 analog multiplexer connected to an Arduino Mega microcontroller, sampling at 200 Hz and transmitted through serial communication at 230400 baud. A multi-branch neural network architecture was implemented to classify the tactile input signals. The model integrates three complementary processing branches: a one-dimensional convolutional neural network (1D-CNN) analyzing raw temporal signals from the 16-channel sensor array, a graph convolutional network (GCN) encoding the spatial adjacency relationships between sensors, and a fully connected branch processing power spectral density features extracted using the Welch method across multiple frequency bands. The outputs from these branches are fused into a shared classification layer that predicts 17 classes, representing the idle state and the 16 possible Braille button positions. Experimental results demonstrate that the proposed system achieves 89% classification accuracy across the 17 classes. The inclusion of spatial graph modeling significantly improves recognition performance compared with purely temporal models, confirming that mechanical coupling between neighboring sensors provides valuable spatial information for identifying touch locations. Overall, the proposed Braille-based self-powered TENG electronic skin demonstrates a promising approach for enabling accessible tactile communication interfaces for deaf-blind individuals. By combining flexible silicone electronics, triboelectric self-powered sensing, and machine learning-based signal interpretation, the system provides a scalable and affordable platform for future wearable assistive communication technologies."
	Paper 78: Real-time dimensional measurement of hot forged components: a pixel-level analysis **Authors:** ***Muthair Hafeez, Muftooh Siddiqi and David Butler*** **Abstract:** "This paper presents a passive dimensional measurement system for real-time quality control in hot forging operations. The research demonstrates the efficacy of pixel-level measurement techniques in achieving dimensional accuracy under elevated temperature conditions, with measurement errors consistently

16:10-16:30 Sep.2	remaining below 1.0 mm. The study encompasses measurements of cylindrical mild steel samples across temperature ranges from ambient to 1300°C, evaluating both diameter and length dimensions before the forging process. Statistical analysis validates the alignment between the proposed system and reference methods. The findings demonstrate that the proposed pixel-level measurement system achieves measurement accuracy requirements necessary for quality control in industrial forging processes, with consistent results across a wide range of sample sizes and process phases."

Session 2D: Modelling and Simulation

(Venue: C2-Ballroom Right - D)

Chair: Prof. Amgad Salama

Time	Content
15:30-15:50 Sep.2	**Paper 2:** Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation [Online] **Authors:** ***Ahmed Basager and Abdullah Alrabghi*** **Abstract:** Balancing operational requirements and sustainability goals poses a persistent challenge in manufacturing systems, as sustainability-driven initiatives such as recycling and reprocessing often compete with productivity, capacity, and utilization constraints. While recycling can reduce material waste and support environmental objectives, its integration into existing production systems may introduce additional workload, congestion, and inefficiencies if not carefully designed. This creates a trade-off where improving sustainability performance may inadvertently degrade operational efficiency. Consequently, decision-makers require systematic, data-driven tools to evaluate these trade-offs and identify operating policies that align sustainability goals with production performance. To address this challenge, this paper presents a simulation-based case study of a production system. A discrete-event simulation model was developed using Simio, based on detailed shop-floor data and statistically fitted input distributions. The study evaluates recycling scenarios and assesses their impact on key performance indicators, including machine utilization, system throughput, and rejection rates. The results show that recycling rate have a significant effect on both operational efficiency and sustainability objectives. The study demonstrates the value of simulation as a decision-support tool for designing sustainable manufacturing systems without compromising operational performance.
15:50-16:10 Sep.2	**Paper 53:** Study of the influence of operating factors on the thermal operating mode of internal combustion engines **Authors:** ***Aidarali Tulenov, Yermek Baubekov, Baurzhan Shoibekov, Gabit Bakyt and Gabit Bekbolatov*** **Abstract:** "The article discusses the influence of various operational factors on the thermal regime of internal combustion engines, in particular the YaMZ-740 engine, which is important for assessing the service life and reliability of engines. The aim

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	of the study is to quantitatively assess the dependence of the coolant temperature on the ambient air temperature, vehicle load, drag coefficient, and degree of traffic congestion on the route. During the work, statistical data on the operation of KamAZ vehicles in various conditions were collected. Using linear and polynomial approximation methods, dependencies between the coolant temperature and the specified factors were established. It was found that the most significant influence on the thermal regime is exerted by the ambient air temperature, to a lesser extent by the load and rolling resistance, and to the least extent by the traffic congestion on the routes. The results obtained allow taking into account the operational features when planning maintenance, diagnosing the technical condition of the engine, and evaluating its operating modes."
16:10-16:30 Sep.2	**Paper 50:** Strategic Governance of Decentralised Autonomous Supply Chains: Macro-Environmental and Contextual Frictions [Online] **Authors:** ***Funlade Sunmola, Patrick Burgess, Memunat Lami Salihu-Yusuf and Hakeem Omolade Sunmola.*** **Abstract:** "Competitiveness in global manufacturing networks is driven by automated Industry 4.0 execution, but transitioning to Industry 5.0 Decentralised Autonomous Supply Chain (DASC) governance introduces strategic bottlenecks. This transition directly impacts distributed manufacturing networks, additive manufacturing scaling, and multi-tier quality assurance, where traditional centralized oversight fails. Unlike conventional supply chains, DASC governance relies on distributed decision-making authority and multi-tier alignment, necessitating flexible frameworks enabled by distributed ledgers. This paper presents a systematic literature review to map the external boundaries conditioning autonomous network coordination. The identified frictions are analysed through Governance, Convention, and Socio-Technical Systems theories. The results formalise an analytical taxonomy mapping key strategic frictions to their operational impacts, including consortium complexity, legal uncertainty, compliance auditing, data confidentiality, and infrastructure deficits. These findings provide operations managers, systems engineers, and strategic decision-makers with a framework to evaluate network readiness, reduce integration risk, and manage structural power dynamics in decentralized manufacturing networks."

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Session 3A: Remanufacturing - 1 (Venue: C2-Ballroom Right - A) Chair: Prof. Mozafar Saadat	
Time	**Content**
17:00-17:20 Sep.2	**Paper 25:** Recycled Carbon Fiber in Circular Economy Applications: Recent Advances and Developments in Sustainable Composites **Authors:** ***Faraj Alsubaie and Liu Yang*** **Abstract:** This research focuses on developing sustainable composite materials by using recycled carbon fibre (rCF) for various industrial purposes. The main goal is to explore how recycled carbon fibre can replace or partly replace new (virgin) carbon fibre in different types of composites, helping to reduce waste and support more environmentally friendly manufacturing. The study is planned in several stages. First, the basic properties of recycled carbon fibre will be carefully compared with new carbon fibre using tests such as fibre strength measurements at different lengths, surface analysis, and other standard material characterization methods. Next, the recycled fibres will be processed into three main types of composites.
17:20-17:40 Sep.2	**Paper 48:** Agentic Generative AI for Aviation MRO Scheduling: A Capability-Gated Autonomy Framework **Authors:** ***Maryam Busakhar and Ali Alsayegh*** **Abstract:** Maintenance planning and scheduling is one of the most demanding decisions in aircraft maintenance, repair and overhaul (MRO), because task durations are uncertain, findings emerge during inspection, and aircraft-on-ground (AOG) pressure competes with airworthiness obligations. Although The problem has been addressed extensively through operations-research and reinforcement-learning methods, whereas agentic generative AI, in which language-model agents plan and re-plan under disruption, has so far been demonstrated mainly in general manufacturing and maintenance decision support outside aviation. This paper examines the adoption of agentic generative AI for MRO scheduling as a process innovation through an innovation-management lens. A critical synthesis integrates three themes: maintenance scheduling in aviation MRO, agentic generative AI for planning and scheduling, and AI adoption as an organisational capability. Within the defined search boundary, no study was identified that applies agentic generative AI directly to aviation MRO scheduling. Existing agentic and generative AI applications in maintenance instead focus mainly on predictive maintenance, retrieval and documentation support, without linking autonomy to demonstrated organisational capability. The paper therefore proposes a capability-gated autonomy framework that links the scheduling autonomy granted to an agent to the organisation's absorptive capacity, dynamic capabilities and task criticality across three tiers with associated controls. The framework is illustrated through a disruption-driven rescheduling scenario, and testable propositions are offered for

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	future empirical evaluation.
17:40-18:00 Sep.2	**Paper 75:** Feasibility of flax/PLA thermoplastic composite for sustainable wind turbine blade repair **Authors:** ***Edward Cant*** **Abstract:** The decommissioning of glass and carbon reinforced thermoset wind turbine blades represents a rapidly growing global waste stream, projected to reach 43 megatonnes by 2050. This study examines flax fibre reinforced polylactic acid (flax/PLA) as a biomass-derived alternative for a 13 m wind turbine blade based on the legacy Vestas V27. A flax/PLA blade was designed comparable with the V27 reference, yielding a power coefficient of 46.3 %. A Blade Element Momentum Theory study was conducted through finite element analysis in ABAQUS and the blade was evaluated under rated (14 m/s), cut-off (25 m/s) and survival (56 m/s) wind conditions. The blade remained within tensile and shear stress limits at rated speed, approached the shear limit at cut-off, and exceeded both limits and acceptable deflection at survival speed. While unsuitable under extreme wind conditions, the results indicate that bulk flax/PLA could withstand typical service conditions, supporting its application as an alternative material for localised blade repair or inclusion within blade construction.

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Session 3B: Advanced Manufacturing Systems (Venue: C2-Ballroom Right - B) Chair: Prof Kostas Salonitis	
Time	**Content**
17:00-17:20 Sep.2	**Paper 21:** Marjan-R1: Design, Fabrication, Thrust Allocation, and Initial Hydrodynamic Parameterization of a Custom Fully-Actuated ROV **Authors:** ***Md Tarique Bin Hamid, Sarvat M. Ahmad, Masroor Khan, Syed Imtiaz Ali Shah and Ali Albeladi*** **Abstract:** This paper presents Marjan-R1, a custom in-house open-frame inspection ROV developed using CAD, CNC machining, and additive manufacturing. The platform uses eight identical vectored thrusters to achieve near-isotropic control authority across all six degrees of freedom. A thrust-to-wrench mapping is derived from measured thruster poses, and singular-value–based metrics are used to quantify translational and rotational isotropy, with comparisons against the open-source BlueROV2 Heavy benchmark. To enable early controller development before full CFD and experimental system identification (SI), we initialize a compact 6-DOF hydrodynamic model using a hybrid workflow: projected areas are obtained from CAD orthographic silhouettes, translational added mass is estimated using the DNV rectangular-plate method, and the remaining damping and rotational terms are seeded via similarity scaling from published parameters. MATLAB simulations evaluate thrust-allocation-based station keeping under a 0.3 m/s surge current while explicitly enforcing T200 saturation, and a disturbance-rejection margin analysis quantifies the available thrust headroom.
17:20-17:40 Sep.2	**Paper 43:** A.C.T.I.V.E. — advanced cybernetic technology for intervention, versatility, and exploration: a rocker-bogie mecanum rover with Kalman-fused localisation, Bézier-curve navigation, and Limelight vision tracking **Authors:** ***Targyn Faizulla and Rustam Askaruly*** **Abstract:** Autonomous mobile robots (AMRs) deployed in hazardous industrial environments must combine structural robustness, smooth trajec- tory execution, and accurate localisation where GPS is unavailable. This paper presents A.C.T.I.V.E., a holonomic rover on a rocker-bogie suspen- sion chassis with mecanum wheels, with three primary contributions. First, the motor-wheel bracket was redesigned via FEA and generative topology optimisation: peak von

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	Mises stress fell by 88 % (71.7 MPa →8.68 MPa), deflection by 98.7 % (37.1 mm →0.47 mm), and the minimum safety factor rose from 0.321 to 2.651 (×8.3), eliminating wheel-camber drift. Second, a Local Pathing framework parameterises trajectories as cubic Bézier curves with Kalman-filtered dead-wheel odometry, holding position RMSE below 0.03 m over 50 m without any external reference. Third, a Limelight vi- sion coprocessor drives joint-level PIDF control of a five-DOF manipulator arm (SAR230 linear extension stage plus four-axis rotational arm), achieving successful grasps across all trials.
17:40-18:00 Sep.2	**Paper 72:** Design and Integration of a Crank-Angle-Synchronised In-Cylinder Gas Sampling System for Heavy-Duty Diesel Engine Combustion Diagnostics **Authors:** ***Hameed Alshammari*** **Abstract:** Understanding the evolution of combustion chemistry within heavy-duty compression ignition engines is essential for improving engine efficiency and reducing pollutant emissions while facilitating the adoption of renewable and low-carbon fuels. Conventional exhaust gas measurements provide only the final products of combustion and do not capture the transient formation of intermediate species inside the combustion chamber. This paper presents the conceptual design and preliminary integration of a crank-angle-synchronised in-cylinder gas sampling system for heavy-duty diesel engine combustion diagnostics. The work focuses on the development of the experimental methodology and the integration strategy required to enable time-resolved in-cylinder gas sampling. The proposed approach combines crank-angle sensing, combustion pressure measurement, synchronised electronic triggering and a high-speed gas sampling valve integrated within a modified cylinder head. A dedicated sampling port has been designed through computer-aided design (CAD) modelling and exploratory machining of sacrificial cylinder heads to assess geometric feasibility while preserving critical engine components. The resulting architecture establishes the foundation for future commissioning and validation of the complete sampling system and provides a platform for investigating combustion chemistry, pollutant precursor formation and renewable fuel combustion in heavy-duty diesel engines.

Session 3C: Smart Manufacturing

(Venue: C2-Ballroom Right - C)

Chair: Prof. Didier Talamona

Time	Content

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17:40-18:00 Sep.2	**Paper 74:** Development of a Low-Cost Intelligent Robotic Waste Sorting System Based on Computer Vision and Autonomous Manipulation **Authors:** ***Dauren Baltabay*** **Abstract:** Recycling technologies have accelerated the development of intelligent robotic systems capable of automating waste sorting operations. Although industrial robotic sorting solutions have demonstrated high efficiency, their widespread implementation remains limited due to high installation and maintenance costs, complex infrastructure requirements, and limited adaptability to small-scale recycling facilities. Consequently, there is a growing need for affordable and intelligent robotic platforms that integrate computer vision, artificial intelligence, and autonomous manipulation while maintaining acceptable sorting performance. This study presents the development of a low-cost intelligent robotic waste sorting system designed to automatically recognize, classify, and manipulate recyclable waste materials using an integrated hardware and software architecture. The proposed system consists of a custom-designed robotic manipulator, a vision subsystem based on a compact RGB camera, an embedded computing platform, and an artificial intelligence module for object detection and classification. The mechanical structure of the manipulator was developed using three-dimensional computer-aided design techniques with emphasis on lightweight construction, modularity, ease of manufacturing through additive technologies, and low production cost. Kinematic modeling was performed to determine the workspace characteristics, optimize the arrangement of the manipulator links, and ensure accurate positioning of the end-effector during grasping operations. The developed mathematical model enables reliable computation of forward and inverse kinematics required for autonomous manipulation of waste objects with different sizes and geometries. The perception subsystem employs a deep learning-based computer vision algorithm capable of detecting multiple categories of recyclable waste, including plastic, paper, metal, and glass, under varying environmental conditions. Following object recognition, the system estimates the target location within the robot workspace and transfers the corresponding coordinates to the motion planning module. Autonomous manipulation is then performed by generating optimized trajectories for object approach, grasping, transportation, and placement into designated recycling containers. The proposed control architecture enables continuous interaction between the perception, decision-making, and robotic manipulation modules, providing fully automated execution of the sorting cycle. To validate the proposed approach, both simulation studies and prototype-based experiments were conducted to evaluate the performance of the integrated robotic system. The experimental analysis focused on object detection accuracy, manipulator positioning precision, grasping reliability, sorting success rate, and operational cycle time. The obtained results demonstrate that the proposed low-cost robotic platform achieves stable autonomous operation while significantly reducing hardware complexity and implementation costs compared with conventional industrial waste sorting systems. The modular architecture also provides flexibility for future expansion through additional sensors, improved end-effectors, advanced motion planning algorithms, and more sophisticated artificial intelligence models. The proposed research contributes to the development of economically accessible intelligent recycling technologies by combining robotic manipulation, computer vision, and autonomous control within a unified robotic platform. The presented

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	system demonstrates the feasibility of implementing affordable AI-assisted robotic sorting solutions for educational laboratories, research institutions, small-scale recycling facilities, and smart waste management infrastructures. Furthermore, the developed platform establishes a flexible foundation for future investigations into adaptive robotic grasping, collaborative robotics, multimodal sensor fusion, digital twin technologies, and autonomous environmental monitoring systems.

Session 3D: Remanufacturing - 2 (Venue: C2-Ballroom Right - D) Chair: Prof. Lee Than	
Time	**Content**
17:00-17:20 Sep.2	**Paper 55:** Development of a hybrid oil heating system in conditions of a switched off diesel engine of a shunting diesel locomotive and analysis of its effectiveness **Authors:** ***Gabit Bakyt, Malik Zhamankulov, Kuantkhan Sarsenov, Berikkhan Tleuzhanov, Irina Ashirbayeva and Aizhan Makhanova*** **Abstract:** The article discusses the problem of increasing the energy efficiency of shunting diesel locomotives in conditions of a significant share of idle work and at low loads. A hybrid engine oil heating system is proposed, which ensures the maintenance of the required temperature regime when the diesel engine is turned off. The developed system is aimed at reducing fuel consumption and reducing emissions of harmful substances. The efficiency of the proposed solution was evaluated taking into account the operating conditions. The results obtained confirm the feasibility of using the hybrid system in shunting.
17:20-17:40 Sep.2	**Paper 60:** Manufacturing Capability Maturity Pathways: A Comparative Study of Emerging Industrial Clusters in Kyrgyzstan, Malaysia and Indonesia **Authors:** ***Lee Lee Than*** **Abstract:** The ability to transition from resource-based production to advanced manufacturing has emerged as a defining determinant of long-term industrial competitiveness, economic resilience, and sustainable development. While resource-rich economies possess significant opportunities for industrial expansion, their development trajectories vary considerably depending on their capacity to transform natural resource advantages into manufacturing capabilities, technological sophistication, and innovation-driven growth. Despite a growing body of literature on industrialization, structural transformation, and economic diversification, limited attention has been paid to the city-level evolution of manufacturing clusters and the mechanisms through which emerging industrial regions progress toward advanced manufacturing and Industry 4.0 ecosystems. This study addresses this gap by proposing a novel Manufacturing Capability Maturity

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	Pathway (MCMP) framework that conceptualizes industrial transformation as a progressive capability-building process across different stages of manufacturing development. The study examines industrial cluster development through a comparative analysis of Kant and Bishkek in Kyrgyzstan, Bekasi and Karawang in Indonesia, and Kulim and Penang in Malaysia. These locations represent distinct stages of manufacturing maturity ranging from resource-based industrial activities to globally integrated, technology-intensive manufacturing ecosystems. The proposed MCMP framework comprises five developmental stages: (1) Resource-Based Production, (2) Manufacturing Emergence, (3) Manufacturing Diversification, (4) Advanced Manufacturing Systems, and (5) Smart Manufacturing and Industry 4.0. By integrating perspectives from industrial cluster theory, manufacturing capability development, and Industry 4.0 transformation, the framework provides a comprehensive lens for understanding regional manufacturing evolution and industrial upgrading. Using a comparative multiple-case study approach, the research evaluates manufacturing capability development across several critical dimensions, including industrial infrastructure, foreign direct investment (FDI), human capital formation, industrial clustering, technological adoption, automation readiness, supply chain integration, innovation capacity, and participation in global production networks. The findings reveal substantial differences in manufacturing maturity among the selected industrial regions. Kant remains predominantly positioned within the resource-based production stage, characterized by limited technological intensity, low-value-added manufacturing activities, and weak integration into global manufacturing networks. Bishkek both in Kyrgyzstan demonstrates a higher level of industrial dynamism and diversification, functioning as an emerging manufacturing and services hub with significant potential for industrial upgrading despite the structural constraints commonly associated with landlocked economies. In contrast, Bekasi and Karawang in Indonesia represent mature manufacturing diversification models supported by strong automotive, electronics, logistics, and export-oriented industrial ecosystems. Kulim and Penang in Malaysia have advanced further toward high-technology manufacturing and innovation-driven development through their capabilities in semiconductor production, electronics manufacturing, research and development, digitalization, automation, and Industry 4.0 implementation. The results suggest that successful industrial transformation depends not solely on manufacturing investment but on the development of integrated industrial ecosystems that simultaneously foster technology adoption, workforce capability enhancement, industrial innovation, logistics connectivity, institutional support, and global value chain participation. Industrial regions exhibiting higher manufacturing maturity consistently demonstrate stronger productivity growth, export competitiveness, innovation performance, and resilience to economic shocks. Furthermore, the transition toward advanced manufacturing is significantly accelerated by strategic investments in digital manufacturing technologies, smart factory systems, industrial automation, advanced materials, manufacturing servitisation, and innovation-oriented industrial policies. This study contributes to both theory and practice by introducing the MCMP framework as a scalable model for assessing and benchmarking manufacturing capability development across emerging industrial regions. This study offers policymakers and industrial stakeholders a strategic roadmap for accelerating industrial upgrading and Industry 4.0 readiness in Kyrgyzstan and other developing middle-income economies. The findings underscore the importance of building manufacturing capabilities as a pathway for

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	transforming resource-dependent regions into competitive, innovative, and sustainable manufacturing ecosystems capable of participating effectively in the global manufacturing landscape.
17:40-18:00 Sep.2	**Paper 4:** Characterization and Modeling of Non-Spherical Granular Materials for High-Temperature Thermal Energy Storage **Authors:** ***Aidana Boribayeva, Luis R. Rojas-Solórzano, Jennifer Curtis, Nicolin Govender and Boris Golman*** **Abstract:** Granular thermal energy storage systems operating at elevated temperatures, such as those used in concentrated solar power (CSP) plants and industrial waste heat recovery, rely on packed or slowly moving beds of particles exposed to temperatures approaching 1000 °C. Under these extreme conditions, heat transfer within the granular bed critically determines storage efficiency, charging and discharging rates, and overall system performance. Despite this importance, numerical modelling of such systems continues to rely predominantly on idealized spherical particles, neglecting the pronounced geometric irregularities of real granular materials and their impact on thermal transport. Recent experimental and numerical studies have shown that non-spherical particle geometries can significantly enhance heat transfer in granular systems by altering packing structure and contact topology. In stationary packed beds, elongated particles formed anisotropic structures with higher coordination numbers and up to 700-fold larger contact areas, resulting in substantially improved conductive heat transfer (Boribayeva et al. [1]). In moving beds, cylindrical particles increased particle-tube contact area by over an order of magnitude, enhancing conductive heat transfer by approximately 12 % and raising tube surface temperatures by more than 120 °C compared to spherical particles (Boribayeva et al. [2]). Building on these findings, this study presents a simulation framework specifically designed to incorporate realistic non-spherical particle shapes into Discrete Element Method (DEM) modelling of high-temperature granular thermal energy storage systems. Calcined bauxite granules for high-temperature TES applications are characterized using high-throughput dynamic imaging. By capturing two-dimensional outlines of individual particles during free-fall, this method minimizes orientation bias. These outlines are subsequently analyzed via Fourier-based multiscale shape descriptors, enabling the quantitative classification of particle morphology across global (macro scale), intermediate (meso scale), and fine (micro scale) geometric features. Based on this classification, representative polyhedral particle models are constructed to accurately preserve angularity, elongation, and surface complexity while remaining computationally tractable for DEM simulations. The reconstructed non-spherical particle assemblies are employed in DEM-based thermal simulations of packed beds and directly compared with equal-volume spherical particle systems under identical geometrical and thermal boundary conditions. Heat transfer performance is evaluated through detailed analysis of packing structure, coordination number and contact area distributions, transient temperature evolution, and effective thermal conductivity of the granular bed. The results demonstrate that incorporating realistic non-spherical particle geometries leads to enhanced thermal performance and more

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	compact packing compared to equal-volume spherical assemblies occupying the same geometric domain, confirming the critical role of particle shape in governing heat transfer in high-temperature granular systems. Beyond performance assessment, this work delivers a transferable modeling workflow, including a database of experimentally derived particle shape descriptors, a library of simulation-ready polyhedral particle models, and a unified workflow linking imaging, characterization, and thermal DEM simulations. The proposed framework provides a practical foundation for the design, optimization, and digital modelling of next-generation granular thermal energy storage systems for sustainable energy and high-temperature manufacturing applications. References: 1. Boribayeva, A., Gvozdeva, X., & Golman, B. (2024). Packing characteristics and heat transfer performance of non-spherical particles for concentrated solar power applications. Energies, 17(18), 4552. 2. Boribayeva, A., Musahanov, A., Baigarina, A., Lukmanov, I., Sultaniyar, S., Patkhollayev, A., Govender, N. & Golman, B. (2025). Experimental and numerical study of non-spherical particles in moving bed heat exchangers. Chemical Engineering Journal Advances, 100859.

Session 4A: Digital Modelling and Monitoring

(Venue: C2-Ballroom Right - A)

Chair: Prof. David Butler

Time	Content
14:00-14:30 Sep.3	**Paper 32:** Real-Time Localized Digital Twin for Quadrotor UAV Flight Monitoring **Authors:** ***Azim Dukenderov, Meirzhan Mazhenov, Alikhan Duisenbay, Md. Hazrat Ali and Essam Shehab*** **Abstract:** The use of the Quadrotor Unmanned Aerial Vehicles (UAVs) in dense urban and industrial environments has raised the need for safe and monitored flight operations. For that purpose, Digital Twin (DT) technology could be used for real-time monitoring and analysis. DT is a virtual replica of a physical system that mirrors its behavior in real-time. This paper proposes a lightweight and localized digital twin framework for standard multi-rotors using easily available hardware. The system creates a wireless Wi-Fi connection between an onboard microcontroller, Arduino Uno R4, and an MPU6050 inertial measurement unit (IMU) and Ublox NEO 6M GPS module in order to broadcast real-time data to a local 3D visualization engine in Unity. The framework has been validated through real flight tests in a cage environment and is able to visualize the actual orientation parameters or attitude angles (roll, pitch, and yaw), position, and plot the vibration profile. Experimental results show low-latency state mirroring of active flight dynamics with mechanical anomaly detection to ensure the operational flight safety of quadrotor UAV.

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14:20-14:40 Sep.3	**Paper 38:** Allowance-Constrained CAD-to-Point-Cloud Registration for Robotic Machining-Margin Analysis of Aircraft Panel Blanks **Authors:** ***Yan Yang, Zhongwei Li, Kai Zhong and Shi Yusheng*** **Abstract:** Automatic CAD-to-point-cloud registration is commonly used before machining allowance evaluation, but an ICP-based best-fit pose may still produce negative allowance and uneven stock distribution. This paper proposes a post-ICP allowance-constrained pose micro-adjustment method for robotic measurement and laser marking of air-rudder panel blanks. Starting from the automatic registration matrices exported by inspection software, signed point-to-CAD allowances are computed, the effect of a small 6-DOF rigid correction is linearized, and a composite objective is minimized. The objective combines low/negative allowance penalties, allowance uniformity and pose regularization. A measured air-rudder panel blank was used for validation. The selected point cloud contains 304,567 valid points, and the CAD model was represented by a triangulated mesh converted from IGES. Compared with the ICP-based automatic registration result, the proposed adjustment reduced the global negative allowance ratio from 10.83% to 2.48%, increased the fifth percentile allowance from -0.47 mm to 0.38 mm, and reduced the main-region allowance standard deviation from 4.39 mm to 3.81 mm. The results show that allowance-aware pose correction can improve manufacturability without replacing the existing automatic registration workflow.
14:40-15:00 Sep.3	**Paper 57:** Development of an Intelligent Control System for an Integrated Low-Carbon Energy System Based on FPGA, ESP32, MQTT Protocol, and a Digital Twin **Authors:** ***Zhazira Shermantayeva*** **Abstract:** The development of low-carbon energy requires intelligent control systems capable of coordinating, in real time, the operation of renewable energy sources, modular nuclear power plants, battery energy storage systems, and Power-to-Gas facilities. This article offered an intelligent control system for an integrated energy system based on an FPGA, an ESP32 microcontroller, the MQTT protocol, and Digital Twin technology. The FPGA enables real-time algorithms, including MPPT, energy-flow management, and protection functions, while the ESP32 performs data acquisition and exchanges information with the digital twin via MQTT. Mathematical and simulation modeling in MATLAB/Simulink was carried out to assess the effectiveness of the developed architecture. The simulation results showed a 27.4% reduction in energy losses, a 41% increase production, and improved overall control efficiency of the integrated energy system. The proposed architecture can be used to develop intelligent low-carbon energy complexes and digital energy platforms.

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Session 4B: Supply Chain for Manufacturing (Venue: C2-Ballroom Right - B) Chair: Prof Dmitry Sizdov	
Time	**Content**
14:00-14:20 Sep.3	**Paper 17:** Effervescent Tablet-Based Nanofluid with Enhanced Thermal Characteristics and Neutron-Absorbing for Rapid Nuclear Reactors LOCA Response **Authors:** ***Naser Ali and Fatima Safar*** **Abstract:** This work investigates an effervescent tablet-based nanofluid that was tailored for enhanced thermal management and higher neutron absorption capability for water-cooled nuclear reactor loss-of-coolant accident (LOCA) scenario. The tablet used contained a total of 4.5 g of mixed and consolidated nanoparticles, effervescent agents, and surfactants. Precisely, 9.22 wt.% polyvinylpyrrolidone (PVP) K30, 2.88 wt.% sodium citrate, 4.67 wt.% aluminum oxide, 0.52 wt.% graphene nanoplatelet, 0.36 wt.% gadolinium oxide, 1.36 wt.% silica-coated boron carbide, 23.65 wt.% citric acid, and 47.34 wt.% sodium bicarbonate. Next, the tablets were immersed in 4 wt.% of boric acid (H_3BO_3) solution to form the suspension. The dispersion stability, thermophysical properties, macroscopic neutron absorption, and thermal performance of the as-prepared nanofluid was then determined and compared to the basefluid. The results have shown that the formed nanofluid exhibits a 15.3%-20.7% and 15.2%-61.8% increase in thermal conductivity and viscosity, respectively. It also showed an increase in the macroscopic neutron absorption by 15% over conventional borated water. These findings along with the adopted approach can provide decision and policy makers as well as the industry a cost-effective and equipment free solution for nuclear reactors LOCA safety management.
14:20-14:40 Sep.3	**Paper 27:** Methane Concentration Forecasting in Underground Mines Using Sensor Data: A Systematic Review of Methods, Datasets, and Open Challenges **Authors:** ***Iskander Akhmetov and Bakyt Kurmanov*** **Abstract:** Background: Methane explosions remain a leading cause of fatalities in underground coal mining. Early warning systems based on sensor data can provide critical lead time for preventive actions, yet no comprehensive survey systematically compares the methods, metrics, and datasets used for methane concentration forecasting. Objective: This review addresses four research questions: (RQ1) which ML/DL methods have been applied and with what performance? (RQ2) what is the role of spatio-temporal modeling? (RQ3) how are class imbalance, sensor drift, and robustness handled? (RQ4) what are the key research gaps? Methods: A systematic literature search was conducted across Scopus, Web of Science, IEEE Xplore, and

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	Google Scholar. We additionally perform an original quantitative analysis of the Mendeley methane dataset (9.2M records, 28 sensors, 106 days). Results: We find that: (1) the warning threshold exceedance rate is extremely low (0.2–1.25% of time), creating severe class imbalance (1:37 to 1:137); (2) median event duration is only 4–8 seconds with strong clustering; (3) autocorrelation at 1-hour lag remains +0.67, making persistence baselines hard to beat; (4) spatio-temporal approaches show promise but only 2–3 mine-specific GNN publications exist. Conclusions: We identify seven data-driven research gaps and recommend classification-based formulations with event-focused metrics over pure regression approaches.
14:40-15:00 Sep.3	**Paper 1:** Agile methodologies in procurement contracts for improving new product development in complex engineering systems. **Authors:** ***Ayaulym Armanova*** **Abstract:** This systematic literature review synthesises the approaches for enhancing agility of the supply chain through procurement contracts. It examines how adaptive contracting, iterative collaboration, and flexible risk-sharing influence responsiveness, performance and innovation in rapid development environments. The findings highlight key contractual mechanisms that support coordination and adaptability in supply chains.

Session 4C: Smart Systems

(Venue: C2-Ballroom Right - C)

Chair: Prof. Sherif Gouda

Time	Content
14:40-15:00 Sep.3	**Paper 23:** Self-Powered Triboelectric Electronic Skin with Braille-Based Interface for Assistive Communication in Deaf–Blind Individuals **Authors:** ***Raiymbek Nurtay, Olzhas Kanatbek, Shabnam Yavari and Gulnur Kalimuldina*** **Abstract:** Communication remains a major challenge for individuals who simultaneously experience visual and hearing impairments. Most existing assistive communication systems rely on visual displays or auditory feedback, limiting their accessibility for deaf-blind users. To address this challenge, this study presents a self-powered electronic skin (e-skin) communication interface based on Braille language patterns, enabling tactile interaction through dot-based input while integrating triboelectric nanogenerator (TENG) sensing for signal generation and recognition. The proposed e-skin was designed as a flexible tactile interface fabricated using silicone 3D printing technology. All structural components were first designed using SolidWorks CAD software and then fabricated using an S300 silicone 3D printer (Fig 1a). The sensing elements consist of triboelectric nanogenerators, where copper foil electrodes are embedded between two layers of

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	printed silicone. To ensure stable bonding between silicone and copper electrodes, a two-component Ecoflex 00-30 elastomer was used during assembly. The final device structure was thermally cured to obtain a flexible, durable, and wearable electronic skin (Fig 1b, 1c). The upper silicone layer of the e-skin incorporates Braille-style tactile dots arranged in a 3 × 2 matrix, representing the Braille writing system used by visually impaired individuals (Fig 1b). By touching or pressing the raised Braille dots, users generate mechanical contact that activates the triboelectric effect between the silicone layers and copper electrodes. As a result, the system produces electrical signals without requiring an external power source (Fig 1d). This self-powered sensing capability enables the device to operate using the mechanical energy generated by finger interaction, making the interface suitable for wearable assistive communication. To interpret the tactile input patterns, a machine-learning-based signal recognition framework was developed. The device integrates an array of 16 independent TENG sensors, each corresponding to a specific spatial location on the Braille interface. Signals are acquired using a CD74HC4067 analog multiplexer connected to an Arduino Mega microcontroller, sampling at 200 Hz and transmitted through serial communication at 230400 baud. A multi-branch neural network architecture was implemented to classify the tactile input signals. The model integrates three complementary processing branches: a one-dimensional convolutional neural network (1D-CNN) analyzing raw temporal signals from the 16-channel sensor array, a graph convolutional network (GCN) encoding the spatial adjacency relationships between sensors, and a fully connected branch processing power spectral density features extracted using the Welch method across multiple frequency bands. The outputs from these branches are fused into a shared classification layer that predicts 17 classes, representing the idle state and the 16 possible Braille button positions. Experimental results demonstrate that the proposed system achieves 89% classification accuracy across the 17 classes. The inclusion of spatial graph modeling significantly improves recognition performance compared with purely temporal models, confirming that mechanical coupling between neighboring sensors provides valuable spatial information for identifying touch locations. Overall, the proposed Braille-based self-powered TENG electronic skin demonstrates a promising approach for enabling accessible tactile communication interfaces for deaf-blind individuals. By combining flexible silicone electronics, triboelectric self-powered sensing, and machine learning-based signal interpretation, the system provides a scalable and affordable platform for future wearable assistive communication technologies.

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Session 4D: Digital Modelling and Monitoring

(Venue: C2-Ballroom Right - D)

Chair: Prof. Mozafar Saadat

Time	Content
14:00-14:20 Sep.3	**Paper 40:** CommandNet for Target Pursuit: Learning-Based Control Evaluation **Authors:** ***Almat Abdimalik, Nurtay Kaziyev and Ilyas Muhammad*** **Abstract:** Reliable target pursuit is an important capability for mobile robots operating in manufacturing, inspection, and service environments. Many reinforcement learning approaches attempt to train complete locomotion controllers or end-to-end policies that map observations directly to robot actions. Although such methods can achieve strong performance, they often require large computational resources, long training times, and careful tuning, especially for legged robots with many actuated joints. This work investigates a modular and computationally efficient alternative for target pursuit with a quadruped robot in NVIDIA Isaac Lab. The proposed framework reuses a frozen pre-trained locomotion policy for the Unitree Go2 robot and adds a learned high-level command-generation network, referred to as CommandNet, on top of it. Instead of controlling the robot joints directly, CommandNet predicts only two base velocity commands: forward velocity (v) and yaw rate (w). These commands are then executed by the frozen locomotion policy, which handles stable quadruped walking. As a result, the learning problem is reduced from full locomotion control to high-level pursuit command generation. CommandNet is first warm-started using demonstrations from a heuristic target-pursuit controller and then fine-tuned using reinforcement learning. The method is evaluated against the original heuristic baseline under deterministic conditions without domain randomization (DR=0). The evaluation uses target reach rate, episode failure rate, mean steps to reach the target, final target distance, and rolling reach-rate stability as performance indicators. Across 1000 heuristic and 1025 learned-controller evaluation episodes, CommandNet achieved a higher target reach rate, increasing performance from 61.20% to 68.98%. The episode failure rate decreased from 38.80% to 31.02%, while the mean number of steps required to reach the target was slightly reduced from 314.29 to 308.68. The learned controller also reduced the mean final target distance from 0.980 m to 0.865 m. These results indicate that learning only the high-level command-generation layer can improve target-pursuit performance without retraining the full locomotion controller.
14:20-14:40 Sep.3	**Paper 54:** Urban Heat Island Dynamics Around Industrial Zones Using Landsat and MODIS Time Series **Authors:** ***Shamma Albedwawi and Naeema Al Hosani*** **Abstract:** Abstract Purpose: This study investigates the spatial and temporal dynamics of urban heat island (UHI) effects in and around major industrial zones, with the objective of quantifying how manufacturing activity and land use change drive thermal anomalies over a multi-decadal period. The findings aim to provide

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	actionable evidence to inform sustainable manufacturing planning, energy-efficient factory siting, and green industrial zone design strategies. Design/Methodology/Approach: A multi-source remote sensing approach is adopted, integrating Landsat 8 and Landsat 9 thermal infrared imagery (30 m resolution) with Moderate Resolution Imaging Spectroradiometer (MODIS) Land Surface Temperature products (MOD11A1, 1 km resolution) over a 20-year observation window (2005–2024). Land use and land cover (LULC) change is analysed using the European Space Agency (ESA) WorldCover 10 m dataset. Industrial zone boundaries are delineated from OpenStreetMap (OSM) and cross-validated against the Global Human Settlement Layer (GHSL). All data acquisition, spectral index computation, and time-series analysis are carried out within the Google Earth Engine (GEE) cloud platform — using only freely and openly available datasets, ensuring full reproducibility. Key indices computed include land surface temperature (LST), Normalised Difference Vegetation Index (NDVI), and Normalised Difference Built-up Index (NDBI). Spatial correlation analysis and Mann-Kendall trend tests are applied to assess the temporal evolution of UHI intensity in relation to industrial footprint expansion. Expected Findings: The study is expected to reveal a measurable and statistically significant UHI intensification within and immediately surrounding active industrial zones, driven by progressive vegetation loss and impervious surface expansion over the study period. Spatial analysis is anticipated to show a positive correlation between industrial built-up density (NDBI) and elevated LST, and a negative correlation with NDVI, consistent with the documented urban greening–cooling relationship. Temporal trends are expected to demonstrate that periods of rapid industrial development correspond to accelerated UHI growth, with persistent nocturnal warming reflecting heat retention in manufactured materials. These outcomes are intended to support evidence-based decision-making in manufacturing facility planning and industrial estate environmental management. Originality and Limitations: This research is among the first to systematically examine the link between industrial zone morphology and UHI dynamics within a manufacturing research context, positioning earth observation as a practical tool for sustainable production system planning. The methodology relies exclusively on open-access satellite datasets and cloud-based processing, making it fully transferable to any global industrial region without specialised infrastructure. Anticipated limitations include the coarser spatial resolution of MODIS data, which may obscure intra-zone thermal variability at finer scales, and the reliance on community-sourced OSM boundaries, which may be incomplete in some regions. Future work will incorporate higher-resolution Sentinel-2 imagery for refined LULC classification and will seek validation against in-situ meteorological station records where available.
14:40-15:00 Sep.3	**Paper 56:** The Decoupled Information Boundary: Resolving the Capacity Truth Paradox in Distributed Autonomous Shared Manufacturing Organisations **Authors:** ***Funlade Sunmola and George Baryannis*** **Abstract:** Advanced decentralised frameworks, particularly the Dynamic Resource Orchestration Framework (DROF-AIBC), leverage AI and blockchain to automate production across shared enterprises. However, widespread adoption is constrained by the Capacity Truth Paradox. While these orchestrators require transparency, competitive nodes rationally obfuscate capacity data to protect proprietary secrets. This study demonstrates that such obfuscation is a strategic economic response that

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	cripples the coordination mechanisms foundational to DROF-AIBC. To resolve this, we introduce the Decoupled Information Boundary, a multi-layered extension to the DROF-AIBC architecture that isolates public supply-network coordination from private factory-floor execution. By utilising localised edge telemetry to generate binary, Zero-Knowledge Operational Proofs (ZKOP), nodes verify production readiness without exposing raw data. This approach enables safe, trust-minimised orchestration, allowing DROF-AIBC to function effectively within competitive, real-world manufacturing ecosystems.

Session 5A: Artificial Intelligence and ML in Manufacturing

(Venue: C2-Ballroom Right - A)

Chair: Prof. Didier Talamona

Time	Content
15:00-15:20 Sep.3	**Paper 30:** Deep Learning-Based Segmentation and Morphological Characterization of Gamma-Prime Precipitates in Superalloy Microstructures **Authors:** ***Isaac Iwediba and Vassili Vorontsov*** **Abstract:** Abstract. A comprehensive evaluation of the latest deep learning models namely, Attention U-Net, U-Net++, and DeepLabv3+ for single-class precipitates within the COCO framework is presented. By leveraging the Mask R-CNN backbone (Detectron2) trained on COCO-Instance Segmentation, the best performance for segmentation was achieved with Dice values ranging from 0.811 to 0.822 and IoU values up to 0.707. Apart from the segmentation performance, eight different morphological features were extracted from the segmented images namely, area fraction, equivalent diameter, aspect ratio, circularity, perimeter, nearest neighbour spacing, edge density, and number of connected components these can be useful for quantitative analysis of precipitate morphology across different microstructural areas. The trained model can be deployed for inference and metric computation of new images to obtain comprehensive CSV output. The results demonstrate that modern semantic segmentation models are capable of reliably performing automated precipitate morphology measurements.
15:20-15:40 Sep.3	**Paper 52:** A Physics-Informed Machine Learning Framework for Defect Prediction in Directed Energy Deposition Using Simulation, Experimental Validation, and Generative AI **Authors:** ***Yerlik Gabdulla and Muhammad Hassan Tanveer*** **Abstract:** The Directed Energy Deposition (DED) 3D printing process has emerged as a critical technology for manufacturing complex metal components, particularly in aerospace engineering. However, its widespread industrial adoption remains limited due to challenges such as defect formation, material wastage, and process inefficiencies. This study addresses these issues by integrating machine learning with simulation-based and experimental data to enhance defect prediction. Ansys

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	software was utilized to generate data, which was subsequently validated using the DED Meltio M450 printer. Experimental results were further employed to train machine learning models, improving predictive accuracy. To accelerate data generation and reduce computational time, a Generative Adversarial Network (GAN) was introduced and validated. The study focuses on Inconel 718 and SS316, materials widely used in aerospace applications, to ensure the practical relevance of the findings. A user-friendly interface was developed to facilitate process parameter input and real-time defect prediction, while feature importance analysis identified key parameters influencing defect formation. The proposed approach enhances the reliability and efficiency of the DED 3D printing process, contributing to its broader industrial adoption.
15:40-16:00 Sep.3	**Paper 64:** AI-driven voice biomarkers for early detection and continuous monitoring of schizophrenia **Authors:** ***Raghad Al-Sakaji, Firda Rahmadani, Haya Al Jaghoub, Shayma Alshehhi, Sarah Alkindi, Faisal Nawaz and Mecit Can Emre Simsekler*** **Abstract:** Traditional diagnosis and monitoring of schizophrenia rely on clinical interviews and standardized assessments that are subjective, time-consuming, and difficult to scale. Artificial intelligence (AI) applied to speech offers an automated, non-invasive alternative that can detect vocal indicators associated with core symptoms. This paper reviews AI-driven voice biomarkers for the early detection and continuous monitoring of schizophrenia. A systematic literature review was conducted following the PRISMA 2020 framework, and the selected studies were appraised through systems-thinking principles for AI in healthcare. Reviewed studies report promising classification accuracies using acoustic, linguistic, semantic, and multimodal features across several languages, yet are constrained by small sample sizes, limited generalizability, weak explainability, workload effects, and unresolved ethical concerns; most remain at the proof-of-concept stage. We argue that a systems-thinking approach is essential to bridge the gap between technical performance and real-world clinical deployment, ensuring usability, safety, and scalability.

Session 5B: Manufacturing Optimisation

(Venue: C2-Ballroom Right - B)

Chair: Prof. Altay Zhakatayev

Time	Content

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15:00-15:20 Sep.3	**Paper 11:** The Application of Process Mapping and Simulation for Optimising Productivity in Low-Volume Assembly Cells **Authors:** ***Daniel Eaves, Andrew Rees, Andrew Thomas, Christian Griffiths and Balaussa Shaimakhan Azubayeva*** **Abstract:** Modern manufacturing is increasingly defined by mass customisation where product variety and complexity are rising, and low-volume, high-complexity (LVHC) production is becoming more prevalent. However, traditional process optimisation methods, rooted in high-volume, repeatable systems, often fall short in LVHC environments. These contexts are characterised by unpredictable demand, bespoke assemblies, and limited applicability of standard lean tools. This work addresses a critical gap in both industrial practice and academic literature: how to optimise Work-In-Progress (WIP) and inventory control in an LVHC production environment using simulation as a decision-support tool. The study presents a detailed case from a UK-based Off-Highway Heavy Plant manufacturer, focusing on a production cell responsible for assembling a complex subcomponent. A discrete event simulation model was developed to replicate the flow of material across the assembly track. The model was validated using real world throughput and WIP data, providing a reliable baseline for experimentation. A staged experimental programme was conducted to evaluate various WIP control strategies. Findings revealed improved spatial efficiency and significant reductions in WIP. This research makes three core contributions: it introduces a novel, simulation-based methodology tailored for LVHC contexts; it quantifies trade-offs between inventory cost, buffer sizing, and delivery reliability; and it delivers a transferable pilot framework with immediate industrial relevance. In doing so, it demonstrates that simulation can enable lean transformation in LVHC environments.
15:20-15:40 Sep.3	**Paper 71:** A geometric-kinematic feasibility framework for robotic cable clip installation **Authors:** ***Yi Wang and Mozafar Saadat*** **Abstract:** Robotic routing and fixation of flexible cable harnesses remain difficult because each clip insertion changes the physical state of the remaining deformable cable. A sequence that appears feasible in global routing may fail during execution when earlier insertions reduce local slack, restrict the grasping window, or force an unsuitable press-in direction. This paper proposes an analytical geometric-kinematic planning framework for execution-aware cable clip installation. The method represents the cable in material coordinates, updates free cable intervals after each snap-fit insertion and evaluates candidate actions using cascaded feasibility checks on material continuity, operation-window availability, press-axis alignment, and shape risk. A representative five-clip case study compares the proposed policy with fixed material-order and greedy local-selection baselines. The case shows that both baselines can generate nominally reasonable but execution-failing sequences, while the proposed policy completes the installation through a non-monotonic sequence that preserves downstream feasibility. These results show that robotic cable routing should be treated as a coupled sequence-and-execution feasibility problem rather than a pure ordering task. The framework provides a planning-level basis for reducing failed insertions and manual rework in high-density industrial cabling

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15:40-16:00 Sep.3	**Paper 2:** Assessing Operational Impacts of Recycling Strategies Using Stochastic Simulation **Authors:** ***Ahmed Basager and Abdullah Alrabghi*** **Abstract:** Balancing operational requirements and sustainability goals poses a persistent challenge in manufacturing systems, as sustainability-driven initiatives such as recycling and reprocessing often compete with productivity, capacity, and utilization constraints. While recycling can reduce material waste and support environmental objectives, its integration into existing production systems may introduce additional workload, congestion, and inefficiencies if not carefully designed. This creates a trade-off where improving sustainability performance may inadvertently degrade operational efficiency. Consequently, decision-makers require systematic, data-driven tools to evaluate these trade-offs and identify operating policies that align sustainability goals with production performance. To address this challenge, this paper presents a simulation-based case study of a production system. A discrete-event simulation model was developed using Simio, based on detailed shop-floor data and statistically fitted input distributions. The study evaluates recycling scenarios and assesses their impact on key performance indicators, including machine utilization, system throughput, and rejection rates. The results show that recycling rate have a significant effect on both operational efficiency and sustainability objectives. The study demonstrates the value of simulation as a decision-support tool for designing sustainable manufacturing systems without compromising operational performance.

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<table>
<tr><th colspan="2">Session 5D: Artificial Intelligence and ML in Manufacturing
(Venue: C2-Ballroom Right - D)
Chair: Prof. Fadhlan Hafizhelmi Kamaru Zaman</th></tr>
<tr><th>Time</th><th>Content</th></tr>
<tr><td>15:00-15:20
Sep.3</td><td>Paper 19: From Prompts to Performance Rankings: Evaluating Large Language Models as Surrogates for Building Performance Simulation to Rank Residential Designs in Hot Climates
Authors: Abdalrahman Alajmi and Ali Alsayegh
Abstract: Building energy simulation (BES) tools such as ESP-r produce authoritative performance rankings for design alternatives but demand significant computational resources and specialist expertise. This study investigates whether large language models (LLMs) can approximate such rankings from textual configuration descriptions alone. Twenty LLMs spanning seven model families were benchmarked against ESP-r ground truth for 24 residential building configurations under Kuwait's extreme summer conditions (48 °C, 75% RH). Each model received an identical zero-shot prompt and was evaluated using Spearman's rank correlation (ρ) and exact positional match ratio. GPT-4o and Gemini 2.5 Flash achieved perfect agreement (ρ = 1.00, 24/24 exact), and most models reproduced the ESP-r ranking closely, with fourteen of twenty reaching ρ ≥ 0.67, none producing a negative or inverted ordering, and the weakest reaching ρ = 0.33. The ground-truth ranking follows a clear building-physics hierarchy that the strongest models recover from qualitative descriptions alone. Performance nonetheless remained model-dependent, varied more within model families than between them, and was not reliably improved by reasoning-enhanced variants. These results indicate that select frontier LLMs encode sufficient building-physics knowledge for rapid design screening in this hot-climate residential setting, although performance heterogeneity means hybrid LLM-plus-simulation workflows remain essential.</td></tr>
<tr><td>15:20-15:40
Sep.3</td><td>Paper 37: Knowledge management in the age of generative AI: opportunities and challenges for Kuwait's energy sector
Authors: Abdalrahman Alajmi and Ali Alsayegh
Abstract: In the face of volatile oil prices and an accelerating global energy transition, Kuwait's energy sector stands at a crossroads where effective knowledge management (KM) has become essential to competitive and sustainable industrial operations. This review examines the role of KM in fostering innovation, efficiency and informed decision-making within energy organisations, and the tools and technologies that support it. It gives particular attention to how generative AI (GenAI) and large language models (LLMs), through techniques such as retrieval-augmented generation, knowledge graphs and AI agents, are reshaping KM practice.</td></tr>
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	Drawing on Kuwait's national energy initiatives, the paper sets out the principal opportunities and challenges for KM in the country. To our knowledge, it is the first review to connect GenAI-enabled KM with Kuwait's energy sector and Vision 2035; it proposes an integrative framework mapping GenAI techniques to the knowledge management cycle, grounds the framework in evidence from manufacturing deployments, sets out how it should be evaluated, and outlines a research agenda for the continued, technology-enabled growth of the sector.
15:40-16:00 Sep.3	**Paper 65:** From Algorithmic Performance to Clinical Adoption: A Review-Based Taxonomy of Artificial Intelligence Clinical Decision Support Deployment Challenges **Authors:** ***Ahmed Saad, Firda Rahmadani, Khaled Toffaha, Mecit Can Emre Simsekler and Khalid Alzarooni*** **Abstract:** Artificial intelligence-based clinical decision support systems (AI-CDSSs) often achieve strong retrospective performance yet remain difficult to use safely in routine care. A model can discriminate well and still fail if its inputs are unavailable at the time of decision, if its output does not fit the clinical workflow, if clinicians cannot interpret or trust the recommendation, or if performance changes after implementation. We conducted a conceptual narrative review of AI-CDSS deployment literature published between 2015 and 2025 and used thematic content analysis to identify recurring barriers. From this synthesis, we developed a two-layer taxonomy. The foundational safety layer includes requirements that should hold for any deployable system, including data quality, demonstrated clinical utility, explainability, validation, governance, and ongoing monitoring. The translation layer describes how a model output is converted into local clinical action and is shaped by five setting-specific pressures: prediction lead time, workflow fit, human oversight, alert burden, and performance drift. We apply the taxonomy to four acute-care contexts: the ICU, ED, general ward, and trauma care. The comparison shows that the foundational requirements remain constant, but their operational handling differs across settings. In the ICU, the main problem is separating signal from noise; in the ED, it is timing and informative missingness; on the ward, it is false-positive burden and action thresholds; and in trauma care, it is the sequential arrival of information. The taxonomy reframes AI-CDSS deployment as a problem of clinical translation, not model installation alone.

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Session 6A: Machining Processes and Advanced Materials (Venue: C2-Ballroom Right - A) Chair: Prof Asma Perveen	
Time	**Content**
16:30-16:50 Sep.3	**Paper 13:** A Finite Element Study of the Graphene Coated- Bamboo Fibres Reinforced Polyethylene Composites. **Authors:** ***Faraj Alsubaie*** **Abstract:** Polymer-based heat exchangers are gaining attention due to their low cost, lightweight, and corrosion resistance, potentially replacing metallic counterparts. This study simulates graphene oxide (GO)-coated bamboo fiber (BF)-reinforced polyethylene (PE) composites using finite element analysis (FEA) in ANSYS for potential use in polymer heat exchangers. GO coatings at 0.2%, 0.3%, and 0.5% were applied to BF (fixed at 40 wt%) reinforced in PE. Mechanical properties (Young's modulus, ultimate tensile strength (UTS), stress, strain, deformation, safety factor) and thermal properties (conductivity, heat flux) were evaluated. Compared to pure PE-BF, the 0.3% GO composite showed a 116-fold increase in Young's modulus and a 105-fold increase in thermal conductivity. This is the first simulation of its kind, offering insights for fiber-reinforced composites in heat exchanger design.
16:50-17:10 Sep.3	**Paper 51:** Integrated Thermodynamic Modeling and Process Optimization for Microstructure Control in LPBF-Processed CoCrNi–CuZr–Ta Medium-Entropy Alloys (MEAs) **Authors:** ***Clifford Omonini*** **Abstract:** Laser powder bed fusion (LPBF) has become a potent method for producing medium-entropy alloys (MEAs) parts that are characterized by refined microstructures and improved mechanical properties. But nonequilibrium solidification is encouraged by LPBF's high thermal gradients and quick solidification, which leads to elemental microsegregation, interdendritic enrichment, and the development of brittle secondary phases. These effects can greatly increase the range of temperatures at which freezing, make them likely to crack, and make additively manufactured parts less reliable. The MEA CoCrNi-alloy when micro-alloyed with CuZr (0-3%) and Ta (1-5%) for LPBF processing is investigated in this work. Thermo-Calc computational thermodynamics will be used to systematically assess solidification behavior through Scheil simulations before experimental fabrication. According to preliminary findings, liquidus and solidus temperatures exhibit significant composition-dependent differences. At various temperatures of the equi-atomic CoCrNi, the freezing range remained at 38.10C. On the addition of CuZr, the freezing range was averagely at 253.20C for all compositions of CuZr added while the addition of Ta at various compositions, kept the freezing range at 163.40C averagely. Simultaneous addition of CuZr and Ta at various compositions kept the freezing range at a range of 1610C to 2810C. The microstructure is

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	dominated by FCC solid solutions, according to phase fraction analysis, whereas Zr and Ta additions encourage the development of secondary phases like Laves (C14/C15), Ni_7Zr_2, and Ni_3Ta, which are linked to segregation-driven solidification instability. Composition windows with a smaller freezing range and less intermetallic formation will be defined using these thermodynamic insights. High relative density and few defects will then be attained by optimizing the LPBF process parameters, such as laser power, scan speed, and hatch spacing through machine learning. Predicted solidification behavior and as-printed microstructures will be correlated through experimental validation. Establishing a predictive, computation-guided framework for the design of LPBF-processable CoCrNi-based MEAs with enhanced solidification stability and less dependence on trial-and-error alloy development is the overall goal of this work.
17:10-17:30 Sep.3	**Paper 49:** Development of Sustainable Aerated Geopolymer Concrete Using Steel Industry By-products **Authors:** ***Kassym Talgatuly, Chang-Seon Shon, Nazerke Kabylbek, Aigerim Baizhigitova and Saken Sandybay*** **Abstract:** The construction sector now faces a relatively common challenge: providing thermally insulating, lightweight building solutions that are less dependent on energy and natural materials. Autoclaved Aerated Geopolymer Concrete (AAGC) could solve this problem by combining low density and low thermal conductivity through a controlled, predominantly closed-cell pore system, thereby reducing the dead load of high-rise buildings and manufacturing and transportation costs, while reducing the reliance on traditional binders and aggregates, which are either in deficit or require a large amount of energy consumption. Ground Granulated Blast-Furnace Slag (GGBFS) and Basic Oxygen Furnace Slag (BOFS) are commonly used Steel Slag (SS) materials in the production of geopolymers as substitutes for cement and natural aggregates. Research shows that BOFS stockpiles may cause relatively serious long-term environmental impact through the leaching of heavy metals into the soil. Moreover, using GGBFS significantly reduces CO_2 emissions from cement production through fuel savings and eliminates the need to sinter calcareous materials. However, BOFS has a major drawback – the presence of free lime (f-CaO) and free magnesia (f-MgO), which may cause immediate and long-term volume expansion due to delayed hydration. Also, for slag to be used in geopolymer, it should be activated with an alkali solution, typically a mixture of sodium silicate (Na_2SiO_3) and sodium hydroxide (NaOH) or potassium hydroxide (KOH). The scope of this research, then, is to produce sustainable AAGC using GGBFS as the binder, stockpiled BOFS as the aggregate, and an alkali activator with KOH. As noted in previous research, steam curing and autoclaving are two ways to reduce the oxide reactivity of BOFS; thus, these curing methods will also be preferred in this research. Lastly, the effect of different aluminum powder contents, as the main expansion agent, on physical and chemical properties will also be investigated. Mix designs for 6 different mixes were prepared based on the following key parameters: - Material characterization through X-ray fluorescence (XRF), X-ray diffraction (XRD), laser particle size distribution (PSD), and thermogravimetric analysis (TGA). - GGBFS and BOFS were used in 100% replacement of natural binder and aggregate, respectively; - The liquid-to-binder (L/B) ratio was taken as 0.35 and 0.32; - The alkali-to-binder ratio was chosen to be 0.400, and the sodium silicate to KOH ratio was considered to be 2.5 as per

	recommendations from the literature. The concentration of KOH was taken as 10 M. - Different aluminum powder content (0.1%, 0.2%, and 0.3% from binder content) to control foaming characteristics of geopolymer. All samples were cured in a steam chamber at 70°C for 2 hours, then placed in an autoclave at 180 °C for 4 hours. Finally, the samples were left to cure under ambient conditions. All mixtures were then tested for compressive strength (ASTM C109), hardened density (ASTM C642), alkali-silica reaction (ASTM C1260), and thermal conductivity (ASTM C177). Additionally, the microstructural characteristics of the mixtures were investigated using Fourier Transform Infrared Spectroscopy (FTIR) and Scanning Electron Microscopy (SEM) with Energy-Dispersive X-ray Spectroscopy (EDS). The compressive strength at a fixed L/B ratio decreased as the aluminum powder content increased. As expected, using more aluminum powder results in more entrained air voids, which negatively affect the compressive strength of the geopolymer concrete. Also, the 28-day compressive strength decreased in almost all mixtures due to rapid curing (steam curing and autoclaving) and drying-shrinkage, which led to the development of microcracks, as confirmed by SEM analysis. The drop in compressive strength due to microcrack development was also confirmed with decreased bulk density of the corresponding mixes during the curing period. Across all mixes, the apparent density increased over 28 days due to continued solidification after early curing, most likely driven by ongoing gel formation. As expected, with a fixed L/B ratio, mixes with higher aluminum powder content have higher Volume of Permeable Pores (VPP%) and lower bulk density at both ages due to gas formation. Overall, BOFS-0.32-0.1 (indicating BOFS-L/B ratio-aluminum powder content) exhibited the highest compressive strength due to a more stable pore structure and continuous solidification. All mixtures with 0.32 L/B showed expansion less than 10% within 14 days, which satisfies the ASTM C1260 criteria for “innocuous behavior in most cases”, while mixes with L/B 0.35, except the one with 0.1%, did not meet this requirement. A lower L/B ratio created a stiffer matrix, which directly suppressed BOFS swelling. Also, mixes with higher aluminum content exhibited greater expansion, contradicting initial expectations. This behavior can be attributed to the presence of f-CaO and f-MgO, which govern the tendency of BOFS to expand. In more porous structures, these compounds could be more readily exposed to and penetrated by the alkaline environment, thereby promoting expansion. The thermal conductivity ranges from 0.1500 to 0.2000 W/mK across the mixes, which is much lower than that of normal-weight concrete. Regardless of the mix, FTIR showed the presence of C-A-S-H, K-A-S-H, and C-S-H (tobermorite), which was also confirmed by the SEM-EDS analysis.

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Session 6B: Automation, Flexible Assembly, and Sensors

(Venue: C2-Ballroom Right - B)

Chair: Prof. Fadhlan Hafizhelmi Kamaru Zaman

Time	Content
16:30-16:50 Sep.3	**Paper 7:** IoT-Enabled Predictive Maintenance for Industrial Automation Systems **Authors:** ***Alisher Yelseitov, Zhalgas Bolatbayev and Tohid Alizadeh*** **Abstract:** The increasing complexity of industrial automation systems under Industry 4.0 requires intelligent, data- driven maintenance strategies. This work presents an Internet of Things (IoT)-enabled predictive maintenance (PdM) framework designed to detect mechanical failures and pressure anomalies in a modular production line. The system integrates real-time sensor data acquisition via Node-RED and Google Sheets with a multi-task machine learning model, enabling binary classification of mechanical faults and multi-class pressure prediction. The architecture was implemented on a FESTO MPS 500 training platform using a Siemens S7-1200 PLC, with time-series features derived from the motion of pneumatic actuators. Experimental results demonstrated high accuracy in fault detection, while pressure prediction showed moderate generalization performance, indicating room for further refinement. The study highlights a cost-effective and scalable approach for PdM, with real-time visualization through MQTT and a mobile/web dashboard. Key limitations include dataset size and model generalization to varied pressure conditions. Overall, this research contributes a practical solution for enhancing reliability and efficiency in a smart manufacturing environment.
16:50-17:10 Sep.3	**Paper 42:** Sensor-based physical safety system for human-robot collaboration **Authors:** ***Bakbergen Yermekbayev, Kaisar Kozhay, Muhammad Abdullah Niazi, Md Hazrat Ali and Essam Shehab*** **Abstract:** This paper introduces a vision-based physical safety Internet of Things (IoT) architecture for human-robot collaboration. The system continuously monitors operator proximity using cameras and halts robot motion when the operator enters an unsafe zone. Two RGB/PoE cameras are connected via a wired local area network (LAN) to an edge control panel, which estimates the operator's position, calculates the minimum distance to the robot safety envelope, applies deterministic stop and resume logic, and transmits concise commands to the robot controller over Wi-Fi. Unlike cloud-based designs, this approach processes video data locally, reduces wireless bandwidth requirements, and limits the remote IoT interface to dashboard, logging, and maintenance functions. The paper focuses on physical safety and secure communication. The primary contribution is a practical and reliable architecture that supports laboratory prototyping of proximity-based robot stoppage, while maintaining a clear separation among functional components: sensing, decision logic, secure robot command transmission, and cybersecurity of data transmission.

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17:10-17:30 Sep.3	**Paper 14:** Optimization of production rate of WEDM using Taguchi Methodology **Authors:** ***Rashmi Arya and Hari Singh*** **Abstract:** The aim of the present research work is the performance study of the Wire Electric Discharge Machining. Aluminium hybrid composite has been considered as work material to explore its machining on Wire Electric Discharge Machine. The Aluminium 8090 is reinforced with Silicon Nitride (6% weight percentage) and Hexagonal Boron Nitride (3% weight percentage), and is manufactured by combining ball milling and stir casting process. Here, design of experiments has been performed by Taguchi methodology's L27 orthogonal array. The effect of pulse on time, servo voltage, peak current, wire feed, and wire tension was studied on production rate.

Session 6C: Additive and Hybrid Manufacturing - 2

(Venue: C2-Ballroom Right - C)

Chair: Prof. Yerbol Sarbassov

Time	Content
16:30-16:50 Sep.3	**Paper 28:** Rapid Liquid Printing in Clay Medium for Construction-scale Concrete 3D Printing: A Process Development **Authors:** ***Mukhammad Safa Akimbay, Bakbergen Temirzakuly, Essam Shehab and Md. Hazrat Ali*** **Abstract:** Rapid Liquid Printing (RLP) offers a route to fabricate non-planar cementitious geometries by depositing fresh concrete within a carrier medium that supports the material during early curing. This study investigates whether an accessible clay-based medium can be adapted for construction-scale RLP, thereby reducing the need for external supports. The experimental workflow is illustrated in Fig. 1. A parametric circular geometry was first developed in Grasshopper, converted into a three-dimensional toolpath, and exported as G-code. Clay-water carrier media with ratios of 6:4, 7:3, and 8:2, with and without 0.02% plasticizer, were then prepared. A cementitious mixture was extruded through a 29 cm nozzle along the programmed path inside the clay bath, and the recovered specimens were visually assessed for nozzle mobility, closure of the medium, strand stability, and retention of open geometry. The consistency of the carrier medium had a pronounced effect on printing performance. The 6:4 clay-to-water mixtures allowed easy nozzle movement but provided insufficient support, leading to strand sagging and irregular cross-sections. The 8:2 mixtures improved support but left persistent nozzle channels and required greater actuator force. The 7:3 clay-to-water mixture containing 0.02% plasticizer gave the best observed balance, allowing the nozzle to travel while the medium closed more reliably behind it and maintained a straighter deposited strand. The resulting circular specimen retained its holes after removal from the clay, demonstrating the preliminary feasibility of printing unsupported open cementitious geometries within a clay-based medium. These results establish a practical process workflow and identify a promising carrier-medium composition

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	for further development of construction-scale RLP. However, the present findings are preliminary and are based mainly on visual assessment. Future work will quantify yield stress and thixotropic recovery, dimensional accuracy, curing behavior, interlayer adhesion, and mechanical performance to determine the structural reliability of the process.
16:50-17:10 Sep.3	**Paper 29:** Geometry-aware inverse process planning for non-planar extrusion additive manufacturing **Authors:** ***Kaicheng Ruan, Yiwei Weng and Yi Xiong*** **Abstract:** Non-planar extrusion additive manufacturing (AM) can fabricate curved surfaces with improved surface conformity, reduced stair-stepping, and lower dependence on support structures. However, current non-planar workflows usually generate the curved toolpath first and then assign layer height, material flow, and related process settings empirically. The purpose of this study is to develop a geometry-aware inverse process-planning method that links local surface geometry to bead formation and generates feasible parameters for non-planar printing.
17:10-17:30 Sep.3	**Paper 44:** A comparative study of energy-efficient WEDM strategies for better surface integrity for machining of hybrid composite **Authors:** ***Nilesh Kumar, Jatinder Kumar and Rajesh Khanna*** **Abstract:** Wire-EDM is a widely used process for machining difficult-to-machine materials, but it also suffers from poor surface integrity. Multi-pass strategy is one of the methods used for improvement in surface quality. The present study aims to compare the process economy and surface integrity of two distinct WEDM strategies while cutting hybrid metal matrix composite (Al6061-90%/SiC-2.5%/TiB2-7.5%), namely: rough cut at low discharge energy and trim cut at various parametric conditions. Process economy includes electrode wire consumption, total machining time, material wastage and associated expenses, whereas recast layer thickness and surface roughness are used as surface integrity characteristics. Results indicate that hybrid strategy is cheaper if it is undertaken after rough cut at higher and medium levels of energy discharge condition, whereas it is found to be ineffective if the proceeding rough cut has been undertaken at low energy condition. Hybrid strategy has the potential to cut down the machining cost by 42-44% if the initial rough cut has been undertaken with high or medium energy parametric setting, while maintaining the similar level of surface quality produced through rough cutting at low discharge energy

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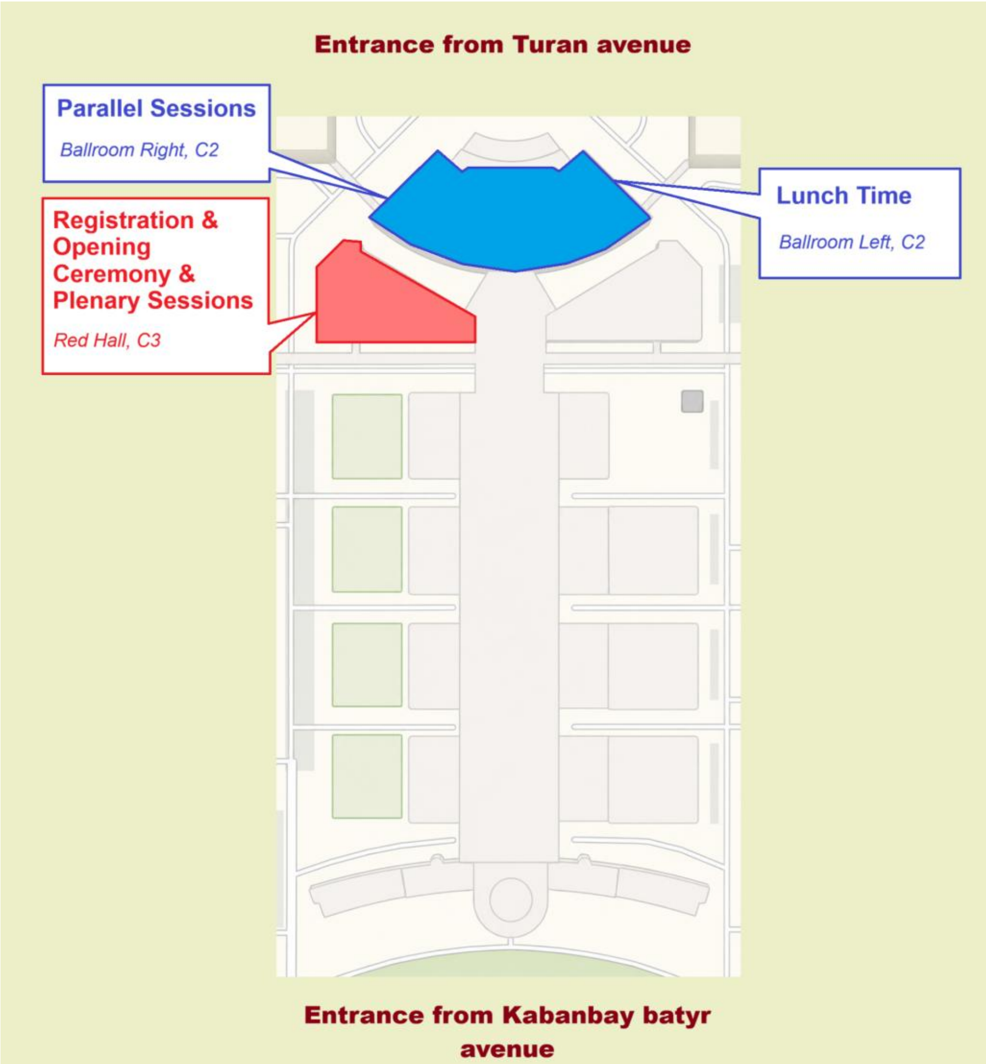
CAMPUS OVERVIEW



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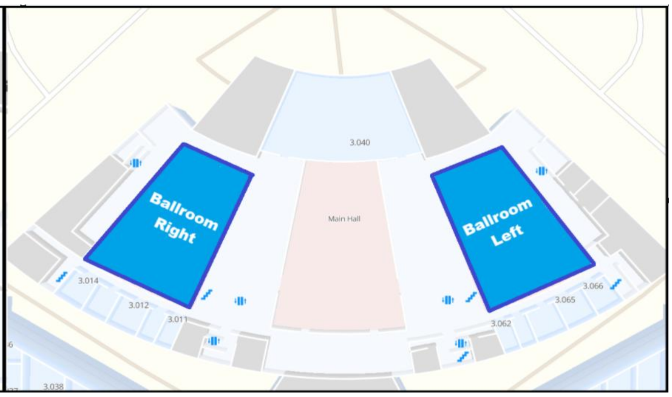

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